

THE SPACE SHUTTLE DECISION

would return a profit to the government by selling 500 of these airliners. Similarly, the funds advanced to Lockheed came from banks, not from the Treasury, with the loan guarantee merely providing security in lieu of collateral. Both ventures, however, were controversial because they opened the door for the government to assume risks that had traditionally rested within the domain of corporate finance, with its banks and securities markets.

In the face of these well-founded objections, the House and Senate nevertheless voted to support Lockheed, even though their members were well aware that they might be setting an unwelcome precedent for further such interventions. In doing this, Congress showed that it would fight the aerospace recession by passing a measure—the Lockheed loan guarantee—that went well beyond the usual demand for pork-barrel spending to provide jobs during hard times. This meant that to support the aerospace industry in its time of difficulty, Washington would go the extra mile and would enact legislation that ordinarily it would not consider. Against this background of industry woes, the Shuttle, which had stirred such controversy during 1970, appeared in 1971 as a straightforward initiative that could win backing on its merits. In this spirit, though continuing to face opposition within the Office of Management and Budget, the shuttle would gain support where it counted most: from Nixon himself.



CHAPTER EIGHT

A Shuttle to Fit the Budget

In May 1971, the Office of Management and Budget (OMB) proposed to limit NASA's spending to a peak of \$3.2 billion during the next five years. George Low, working with his NASA associate Willis Shapley, responded by proposing a constant budget, with shuttle spending rising no higher than a billion dollars per year. Low would later describe this approach as "the only one which could convince the OMB that we could do the shuttle and at the same time have a balanced space program." Unfortunately, this billion-dollar limit left NASA with funds to build an orbiter but not a booster. Could the agency find a way to build both?

The Orbiter: Convergence to a Good Solution

As early as 1969, during the initial Phase A studies, Lockheed had taken the initiative in proposing a two-stage fully-reusable design with both stages built of aluminum and using silica tiles for thermal protection. While the final design for the shuttle orbiter would in fact use this approach, one must not think that Lockheed was prescient. Though that company indeed was in the forefront in developing such tiles, they were items for laboratory research. A design that specified their use had no more intrinsic credibility than one that proposed to use the miracle metals Unobtainium and Wishalloy. Nevertheless, the work at Lockheed suited an emerging preference within the Air Force.

Late in 1969, Air Force officials stated that they wanted to build the orbiter using a conventional aluminum airframe, along with whatever form of

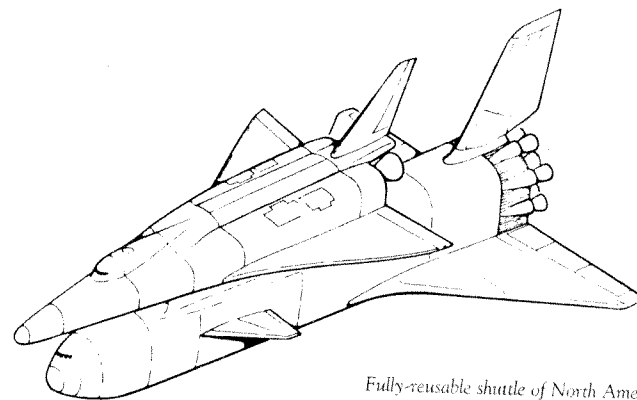
thermal protection would be appropriate. In contrast to strong reliance on titanium in hot structures, this preference for aluminum stemmed from an Air Force finding that the aerospace industry faced a shortage of the specialized machine tools needed to fabricate large structural parts from titanium alloy.

Within NASA and its contractors, design studies weighed the relative merits of aluminum and titanium as primary structural materials. The aluminum airframe promised to be lighter in weight, reflecting the fact that aluminum is lighter than titanium. It also would be less costly to build, reflecting the industry's long experience with aluminum. By contrast, titanium structures promised to cost up to three times as much as their aluminum counterparts, and would carry greater risk in development.

Titanium, however, could overcome these disadvantages with its ability to withstand temperatures of 650 °F, compared with 300 degrees for aluminum. This brought a considerable reduction in the weight of the thermal protection, for two reasons. The temperature resistance of titanium would make it possible to build the top areas of the wing and fuselage of this metal alone, without additional thermal protection, for they would be shielded against the extreme temperatures of re-entry by the bottom of the vehicle. In addition to this, a titanium structure could function as a heat sink, absorbing some heat and thereby reducing the thickness and the effectiveness of thermal protection where it would be needed.

Overall, the advantages of titanium promised a complete orbiter, including thermal protection, that would weigh some fifteen percent less than a counterpart built of aluminum. With the titanium orbiter requiring less thermal protection, it also would cost less to refurbish between missions. Though the higher cost and risk of titanium would militate in favor of aluminum once NASA faced the OMB's cost ceiling, the merits of titanium encouraged its use during NASA's design work of 1970 and 1971.¹

The Phase B studies represented the main line of effort, with John Yardley directing the work at McDonnell Douglas and Bastian Hello managing the activity at North American Rockwell. Both contractors proposed fully-reusable orbiters, carrying all propellant tankage within the fuselage. They looked like large delta-winged fighter planes; indeed, the McDonnell concept



Fully-reusable shuttle of North American Rockwell. (North American Rockwell)

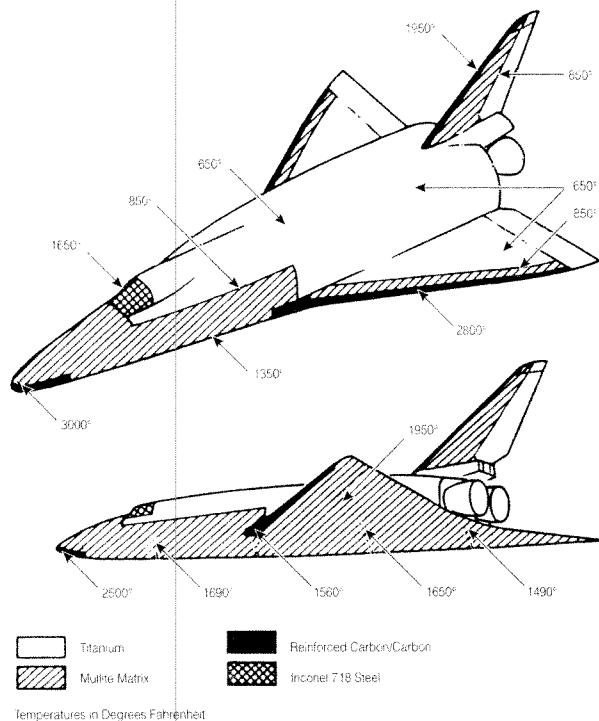
somewhat resembled that company's F-4 Phantom aircraft, which were seeing extensive use in Vietnam.²

Both orbiter concepts called for building the propellant tankage of aluminum. Though the tanks would be empty of propellant during reentry, they would require insulation to keep their contents—liquid hydrogen and liquid oxygen—from evaporating. That same insulation then would protect the aluminum from heat that would soak through the thermal protection. Similarly, the crew compartment was to be of aluminum, with the crew members riding amid an insulated and air-conditioned coolness that would protect the adjacent structure from overheating as well.

In the structural frames and outer skin of the wings and fuselage, the contractors proposed to use titanium freely. They differed, however, in their approaches to thermal protection. McDonnell Douglas continued to favor hot structures, with insulation to protect the underlying framework and temperature-resistant metal panels facing the heat of reentry. Metallurgists had developed specialized alloys of nickel and cobalt for the turbine blades of jet engines; these metals resisted oxidation when hot, making them suitable for

1. *Aviation Week*, January 18, 1971, pp. 36-39; Jenkins, *Space Shuttle*, pp. 67, 95; letter, Low to Professor John Logsdon, January 23, 1979.

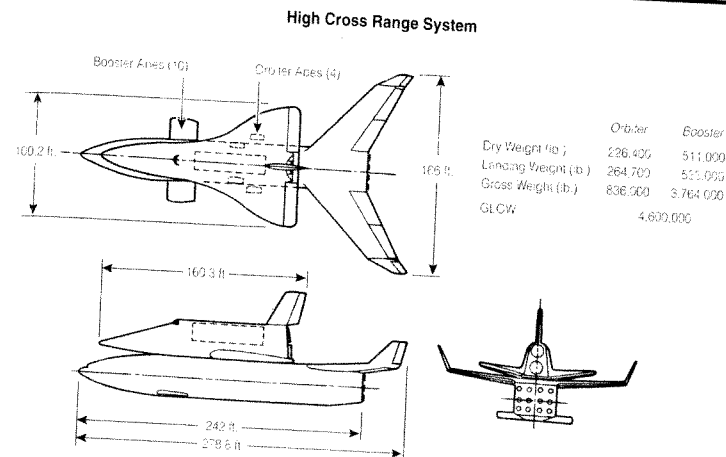
2. *Aviation Week*, April 5, 1971, pp. 36-38; June 7, 1971, pp. 55-61.



Thermal protection on the North American orbiter. (North American Rockwell)

the skin of this orbiter. For instance, most of its underside was to be covered with shingles of Hastelloy-X, a chrome-nickel alloy. The wing leading edges would use struts and beams of columbium; shingles of coated columbium protected areas that were too hot for Hastelloy. Though only a small coterie of engineers had experience with such materials, the McDonnell Douglas designers were not reticent about pushing the state of the art.³

3. AIAA Paper 71-804; Report MDC E0308 (McDonnell Douglas).



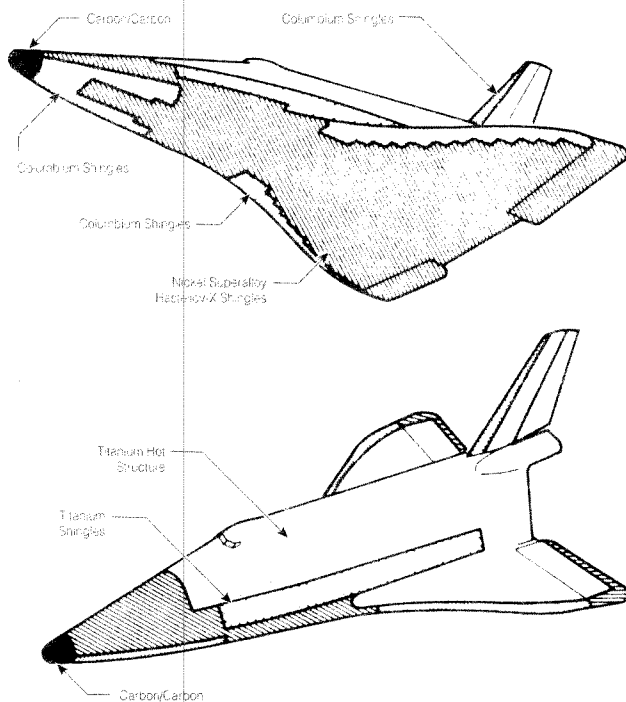
Fully reusable shuttle of McDonnell Douglas. (NASA)

North American also had a strong interest in titanium hot structures, expecting to use them as well on the upper wing surfaces and the upper fuselage. Everywhere else possible, this company's design called for applying thermally-protective tiles directly to a skin of titanium. Such tiles would cover the entire underside of the wings and fuselage, along with much of the fuselage forward of the wings. The main exceptions included the nose and leading edges, protected with carbon composite, and the vertical fin, designed as a hot structure with a skin of Inconel 718, a heat-resistant nickel steel.⁴

These orbiter concepts represented the fruit of several years of experience in the design and study of two-stage fully-reusable shuttles. They also were quite representative of what NASA now would *not* be able to build, under the fiscal limits of the OMB. The agency, however, had an ace in the hole, for in parallel with these Phase B efforts, Lockheed and Grumman had been pursuing studies of alternatives.

4. AIAA Paper 71-805; Reports SV 71-28, SD 71-114-1 (North American Rockwell).

TPS Material Definition



Thermal protection on the McDonnell Douglas orbiter. (McDonnell Douglas)

For both companies, the point of departure lay in partially-reusable configurations that would carry their propellant in expendable tanks. This offered a route to lower development cost because the orbiter could shrink in size by carrying its propellant externally. The tanks could take form as simple aluminum shells, while the orbiter would have much less volume to enclose within its hot structures, and much less surface area to protect thermally.

A Shuttle to Fit the Budget

Such approaches dated to the original Star-Clipper concept of 1965. They had lost favor because the throwaway tankage would cost money, in dollars that literally would be thrown away. Partial reusability was attractive to the OMB because it would cut the all-important development cost, the year-to-year line item in the federal budget. NASA's fully-reusable approach promised huge up-front costs in return for the goal of large savings through reduced cost per flight. The OMB insisted on smaller short-term outlays in exchange for the prospect of a somewhat greater cost of launch, 15 years down the road. The OMB held most of the cards.

Among the alternative shuttle designs, Lockheed was continuing to examine variants of Star-Clipper, with a lifting body for its core vehicle and external tanks flanking this core. The new versions included a two-stage fully-reusable design that placed the propellant tanks within the core and used a McDonnell Douglas shuttle booster as the first stage. NASA by now had little interest in lifting bodies, for they promised difficulty in development along with high landing speeds. With the main line of design activity now defined by delta-winged versions of Max Faget's two-stage rocket airplanes, NASA's continuing support of Lockheed amounted to telling the OMB: Since you want us to look at alternatives, here is what we have.⁵

By contrast, Grumman began its examination of alternative designs with no preconceived views as to how to proceed. When the work began, in mid-1970, this company started with 29 configurations in three categories: two-stage fully reusable, reusable orbiter with expendable booster, and reusable orbiter with expendable propellant tankage. Like everyone else, the Grumman manager, Lawrence Mead, concluded that full two-stage reusability would be best in the long run. His report, however, noted pointedly that the other approaches offered promising means of reducing the peak funding.

Meanwhile, during the fall of 1970, the Grumman group supplemented its NASA funds with company money, to broaden further the range of alternatives. The most promising modified the basic two-stage fully-reusable approach by removing the orbiter's liquid hydrogen fuel from within its fuselage, and storing it in a pair of expendable tanks. Grumman managers presented this concept to officials at the Manned Spacecraft Center in

5. Report LMSC-A989142 (Lockheed); Jenkins, *Space Shuttle*, pp. 98-103.

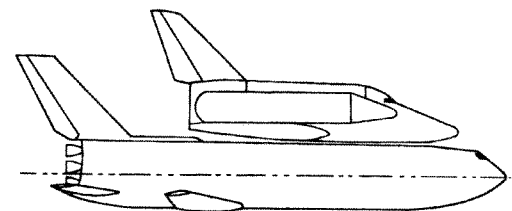
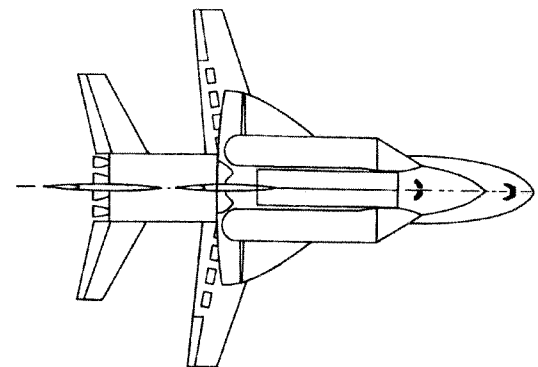
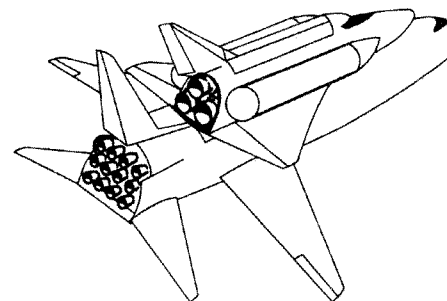
November 1970. Within weeks, these managers were instructed to concentrate their efforts on further and more detailed study of this expendable-tank orbiter, and to compare it with a fully-reusable variant having internal tankage. This approach gained further favor in March 1971, as NASA instructed McDonnell Douglas and North American Rockwell to develop variants of their fully-reusable configurations that would also place the orbiter's liquid hydrogen within expendable tankage.⁶

Why was this approach so promising? Liquid hydrogen is bulky, having only one-fourteenth the density of water. Thus, although it makes up only about one-seventh of a shuttle's propellant load by weight, with six-sevenths being liquid oxygen, liquid hydrogen accounts for nearly three-fourths of the volume. Being low in density and hence light in weight, this fuel could be carried in external tanks of similar light weight. Being bulky, its removal would bring a welcome reduction in the vehicle size and surface area.

In addition to this, Grumman's approach brought a useful shrinkage in the size and weight of the complete two-stage shuttle, including the booster. In designing two-stage vehicles, standard procedures exist for choosing the best staging velocity, so as to achieve the lowest weight of the two stages together. At a higher staging velocity, the first stage becomes excessively large and heavy; at a lower velocity, the orbiter requires more size and weight. Ground rules set at Marshall Space Flight Center, based on such optimization, defined this staging velocity as being close to 10,000 feet per second. For Grumman's two-stage fully-reusable configuration, used as a reference for purpose of comparison, it was 9,750 ft/sec.

With the liquid hydrogen now to be carried in lightweight external tankage, a re-calculation of this optimum showed that it would be advantageous to make these tanks larger, allowing them to carry more of this fuel. The orbiter then would have to carry more liquid oxygen, stored within its fuselage. This, however, would be easy to accommodate. Liquid oxygen is dense, denser than water. Hence, the extra quantity would require little additional volume and would not compromise the overall design of the orbiter.

The upshot was that the optimum staging velocity, concomitant with the lowest overall weight, would drop to only 7,000 ft/sec. This would greatly ease the task of designing the booster. The booster now could be considerably



Grumman's two-stage shuttle, which put the orbiter's liquid hydrogen in expendable tanks.
(Art by Dennis Jenkins)

6. Report B35-43 RP-11 (Grumman), Section 1; *Aviation Week*, March 29, 1971, pp. 45-46.

smaller and lighter in weight, reflecting its reduction in required performance. It also would be much easier to protect thermally.

The orbiter design followed standard practice, with a main structure of titanium. Grumman's report stated that the company would rely on "materials, analysis, manufacturing and test procedures developed for the F-14 aircraft," a Navy fighter that was just entering service. The orbiter's thermal protection was to rely primarily on hot structures, in the fashion of McDonnell Douglas, with carbon composite and silica tiles at the nose and leading edges. The use of external tankage cut the dry or unfueled weight of the complete two-stage shuttle by nearly one-third, from 1.02 million pounds to 692,000 pounds. In the words of the report, this weight saving "means structure we eliminate from design, do not provide tooling for, nor build, maintain, refurbish or otherwise pay for."

This comparison of weights drew on the fact that Grumman had worked to encourage such comparisons deliberately, by carrying through studies of designs with both external and internal orbiter tankage, according to the same ground rules. The report noted that

for those who have in the past undertaken to compare configurations from several contractors, the necessity for a dual effort of this nature is readily apparent. There is nothing more frustrating and inaccurate than to attempt to compare weight, performance, and cost from several contractors, using, by definition, their own unique preliminary design groundrules and criteria.

As a consequence, Grumman's work had to be taken seriously when it pointed to financial advantage:

	Fully Reusable	External Hydrogen
Development cost, \$ M	7,777	6,497
Peak funding, \$ B	2.20	1.85
Cost per flight, \$ M	4.29	4.22

The peak funding level, \$1.85 billion, was a long way from the OMB requirement of \$1 billion. Nevertheless, it was \$350 million closer to this goal than the fully-reusable design. Moreover, in a brilliant example of having one's cake and eating it, Grumman proposed that the expendable tankage

would actually *reduce* the cost per flight. The tanks per se would cost \$740,000 per flight. Other savings, however, would more than offset this, with the largest of them stemming from a substantial cut in the amount of propellants for a flight, and from eliminating the need to refurbish the thermal protection of the now-simpler booster.⁷

While Lockheed's Star-Clipper was widely known for its use of external tankage, it used no booster, relying on a single set of engines in the core to carry it from liftoff to orbit. The use of a reusable booster, in conjunction with expendable tankage on the orbiter, now opened new prospects to explore. An important group of explorers worked at McDonnell Douglas, where they proposed particularly large external tanks that would allow the orbiter to ignite its engines at liftoff, with these engines burning all the way to orbit. Though this shuttle still needed a booster, its staging velocity fell even further, to 6,200 ft/sec. This booster's dry weight, 346,000 pounds, was only three-fifths that of the booster in the fully-reusable system. For the complete two-stage shuttle, cost of development dropped from \$9.82 billion to \$8.67 billion, yielding a reduction on a par with that projected by Grumman.⁸

Another group of designers worked with Max Faget at NASA's Manned Spacecraft Center. Decisions dating to early 1970 had given that center responsibility for technical direction of orbiter concepts, with NASA Marshall receiving similar responsibility for boosters. Faget had responded by initiating studies of a succession of two-stage, fully-reusable configurations, indulging to the full his taste for straight wings and for lightweight payloads in small payload bays that would suit the mission of space-station resupply. In January 1971, when NASA officials had met with Air Force counterparts, they had agreed that the orbiter should have delta wings along with a payload capacity of 60 x 15 feet and 65,000 pounds. Faget, nevertheless, had gone his merry way, as if this agreement did not apply to him. He continued to pursue his personal preferences in shuttle design; it was well into 1971 before he was caught up with the rest of the world.

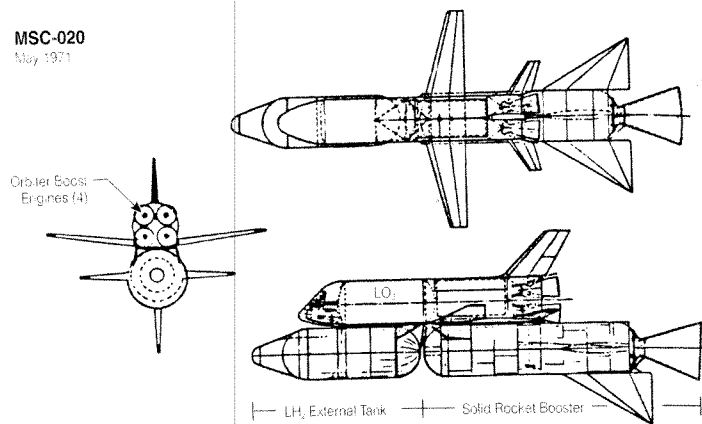
He knew a good thing when he saw it. Although he had strongly advocated the fully-reusable orbiter with internal tankage, he quickly turned to designs with expendable tanks. The Grumman and McDonnell Douglas con-

7. Report B35-43 RP-11 (Grumman); *Aviation Week*, July 12, 1971, pp. 36-39; Powers, *Shuttle*, p. 240.

8. Report MDC E0376-1 (McDonnell Douglas).

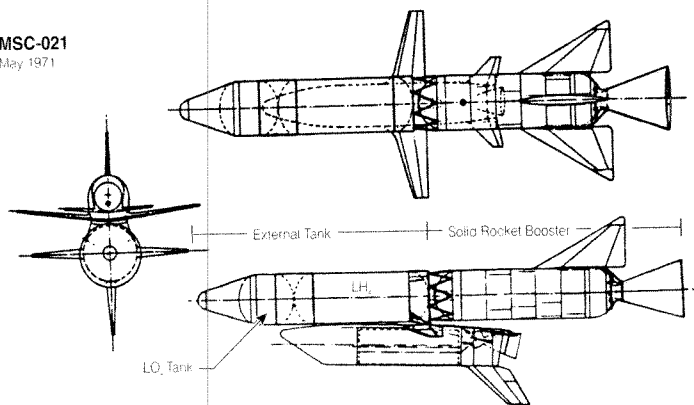
MSC-020

May 1971



MSC-021

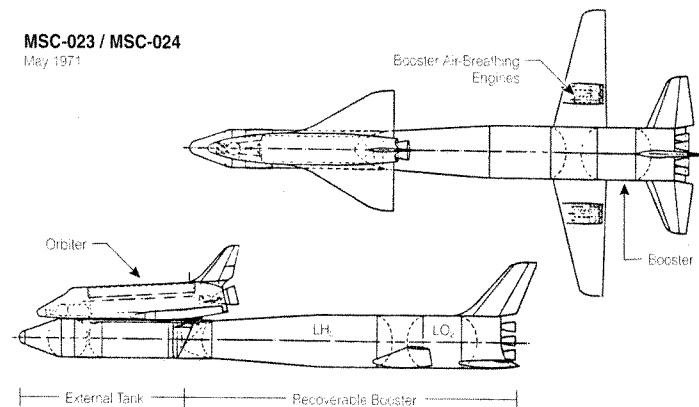
May 1971



How Max Faget adopted external tankage and moved toward the final orbiter concept, 1971.
(Art by Dennis Jenkins)

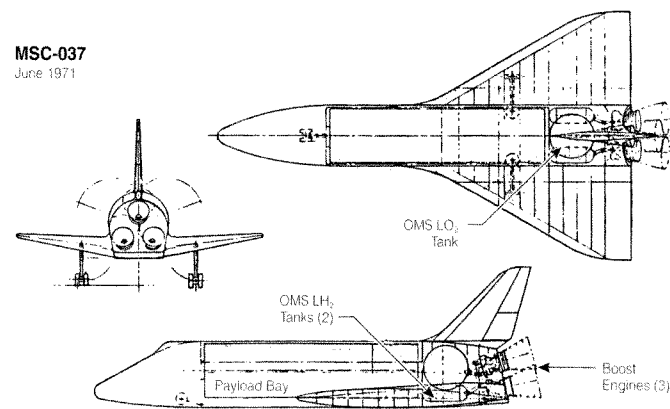
MSC-023 / MSC-024

May 1971



MSC-037

June 1971



cepts had carried their hydrogen in long cylinders running alongside the orbiter fuselage, just above the wing. As early as May 1971, Faget prepared a set of drawings designated MSC-020 that put the hydrogen in a single tank, slung beneath the front of the vehicle. This concept followed his preferences by specifying straight wings, a 20,000-pound payload, and a 30 x 15-foot bay. It showed that, at least in this respect, he was willing to change with the times.

The next step was to lengthen this single external tank to allow it to carry liquid oxygen as well. This would reduce the size of the orbiter to a bare minimum. The tank, attached to the orbiter's belly, would demand structural strengthening, for its store of liquid oxygen would be quite heavy. With all propellant removed from the orbiter, that vehicle could achieve a standard design, independent of the tank. The tank could grow to a particularly large size, further lowering the staging velocity of the booster. In turn, this lower staging velocity would further reduce the size of the booster, cutting the cost of the Shuttle program anew.

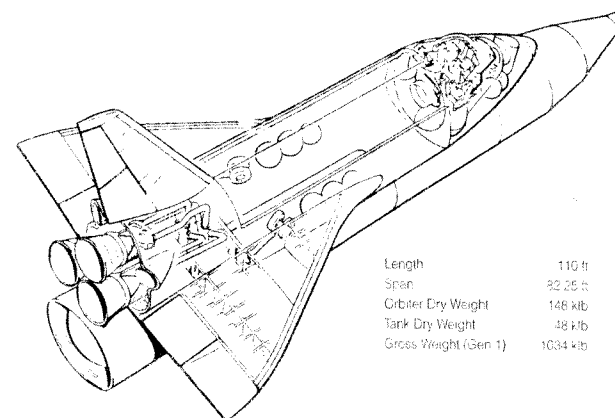
The first such concept, the MSC-021, came forth during that May. Again, it featured straight wings and the same payload weight as in the MSC-020. Though the bay now had a length of 40 feet, it still was much shorter than what the Air Force would accept. During that same month, Faget also proposed the MSC-023, again with all propellants for the orbiter in a single large underbelly tank. It featured delta wings and a full-size bay, 15 x 60 feet. Here, for the first time, was the outline of a shuttle orbiter that would actually be built.

Even so, Faget was not ready to offer an uncritical embrace of delta wings and large payload bays. Though he now had an external tank that he liked and would stick with, his subsequent designs continued to show small bays and straight or swept wings, and sometimes both. He also examined a number of variations in the arrangement of the orbiter's main engines. Though the MSC-023 had only a single such engine, in June he released the MSC-037, amounting to a variant with three engines and a 40,000-pound payload. It matched the final design in important respects.

The contractors quickly followed this lead, as they launched new studies that assessed its merits. Grumman, which initially put only the hydrogen in external tanks, declared that an orbiter such as the MSC-037 was at least as promising. Lockheed, McDonnell Douglas, and North American found that it was superior, with all three firms giving it a strong endorsement. Indeed, as early as September 1, North American presented its own version of the

LO₂ / LH₂ Low Cost Configuration

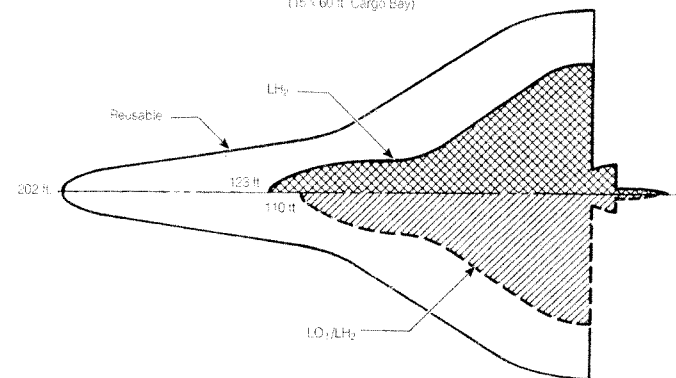
(15 x 60 Cargo Bay)



Orbiter concept of September 1, 1971, which foreshadowed the Shuttle that NASA would build. (North American Rockwell)

LO₂ / LH₂ Orbiter vs. LH₂

(15 x 60 ft Cargo Bay)



Shrinkage of the shuttle orbiter. Planforms compare the sizes of the fully-reusable orbiter with internal tankage; the orbiter with internal LO₂ and with LH₂ in external tanks; and the final orbiter, with all propellants carried externally. (North American Rockwell)

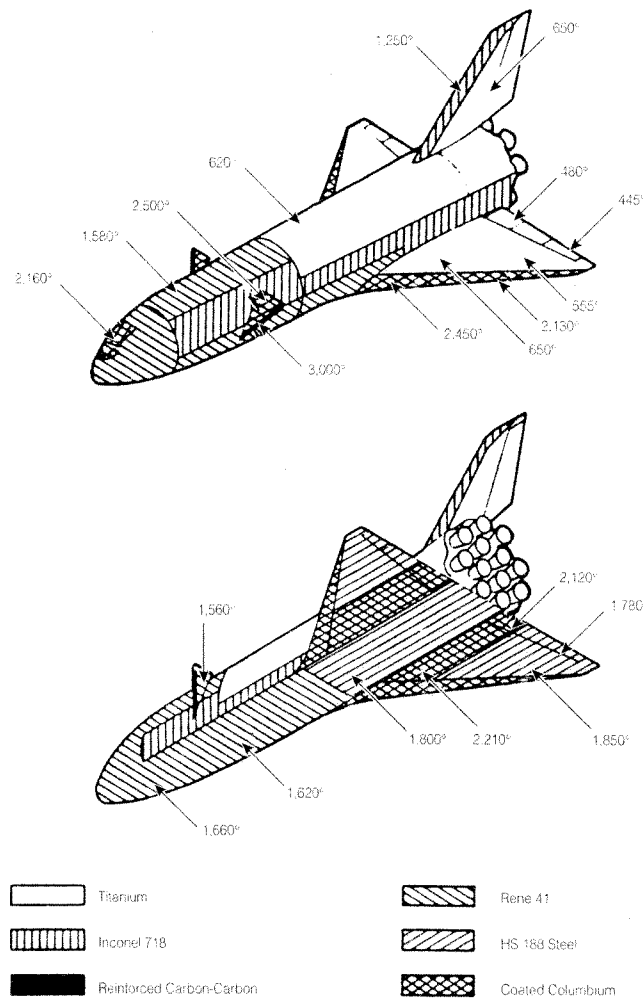
MSC-037, singling out this design as the one to pursue. An artist's rendering showed an orbiter closely resembling the one that NASA would build. Eleven days later, NASA formally instructed its contractors to adopt a variant, the MSC-040, and to use it as a basis for comparison within their ongoing studies.⁹

The Booster: Confusion and Doubt

In contrast to the rapid convergence of the orbiter designs, the booster studies fell into disarray. The initial points of reference lay in the Phase B booster concepts of McDonnell Douglas and North American Rockwell. Both contractors proposed vehicles somewhat larger than a Boeing 747, and weighing some five times as much when fueled. The North American concept had two dozen engines: 12 SSMEs and 12 jet engines, the latter serving during flyback to the launch site. McDonnell Douglas's booster, only slightly less ambitious, was to use a dozen SSMEs as well, with ten turbojets.

The North American orbiter had featured a primary structure and skin of titanium, with tiles providing most of the thermal protection. The booster design was more eclectic. Its front section, which enclosed the crew compartment, had frames and skin of René 41, a nickel-chromium alloy that contained cobalt and molybdenum. Though it was strong and oxidation-resistant, it was hard to fabricate. Wings, vertical fin and canards, small forward-mounted winglets used for control, all were of titanium. The fuselage and its tankage were largely of aluminum.

Then came the thermal protection, which avoided the use of tiles in favor of hot structures. The designers might have simplified matters by specifying the wide use of tiles, as on the orbiter. Tiles, however, were in their infancy and North American had to show that it also understood the design of hot structures. As usual, bare titanium skin sufficed for the upper surfaces of the wings and fuselage. The underside, nose and forward fuselage, however, were a metallurgist's delight—or nightmare. The wing leading edges would use coated columbium. Large surface areas would rely on the alloys René 41 and Haynes 188, which were exotic mixtures of nickel, cobalt, chromium, tungsten, and



Thermal protection on the North American booster. (North American Rockwell)

9. Jenkins, *Space Shuttle*, pp. 107-116; Hallion, *Hypersonic*, pp. 1018, 1048-1068; Reports SV 71-40 (North American Rockwell); (Grimman); LMSC A-995931 (Lockheed), Section 1, *Interim Report* (McDonnell Douglas).

molybdenum. Because these metals would expand when hot, designers proposed to build the thermal protection in the form of shells, free to slide over the underlying structure. On paper, at least, the shells would not come loose.

Such a design certainly was not in keeping with the preferences of Max Faget, who always sought the most conservative approach. It was an engineering efflorescence to match the economic efflorescence of the cost-benefit studies at Mathematica, and was likely to raise just as many eyebrows. Moreover, it showed that the exuberance of 1969, when Tom Paine wanted to go to Mars, was still alive. By contrast, much engineering experience is summed up in the acronym KISS: Keep It Simple, Stupid. This is what John Yardley's designers had done at McDonnell Douglas, in crafting their own booster concept.

One can describe it succinctly: aluminum primary structure throughout, including wings, canards, and tail, with tiles of varying thickness to provide the thermal protection. Though metallic shingles would protect the leading edges, they would see use only in the limited areas where tiles would fail to suffice.¹⁰

These design exercises gave both contractors the opportunity to conduct detailed investigations using a range of approaches: hot structures, aluminum and titanium primary structures, tiles, and metallic thermal protection. Neither contractor offered anything so simple as an aluminum orbiter protected with tiles, though McDonnell Douglas used this approach in its booster. Despite their complexities, however, the concepts were not obviously infeasible, and some people believed they could actually be built.

Along came the Grumman design studies, which put the orbiter's liquid hydrogen in external tanks and lowered the booster's staging velocity from 9,750 to 7,000 ft/sec. Boeing, which had built the first stage of the Saturn V, was teamed with Grumman and had responsibility for the booster. Its engineers determined that this reduction in velocity would bring a disproportionate reduction in the size and weight of the booster, which now would require much less propellant. In addition to this, the lower velocity meant that the booster would be much closer to the launch site when it released the orbiter and began its return. Being closer, it would need less fuel for its jet engines, further reducing its size. The upshot was that whereas the

fully-reusable orbiter's booster would have a dry weight of 798,500 pounds, the new orbiter would cut this booster weight to 494,900 pounds.

Better still, the lower staging velocity virtually eliminated the need for thermal protection. The booster now would need neither tiles nor exotic metals. Instead, it would use its structure as a heat sink, just as with the X-15. During reentry, it would experience a sharp but brief pulse of heat, which a conventional aircraft structure could accept and absorb without exceeding temperature limits. Hot areas would continue to demand a titanium hot structure, which would cover some one-eighth of the booster. The rest of this vehicle, however, would make considerable use of aluminum.

How could bare aluminum, without thermal protection, serve in a shuttle booster? It was common understanding that aluminum airframes would lose strength due to aerodynamic heating at speeds beyond Mach 2; higher speeds required titanium, with its greater temperature resistance. These principles, however, dealt with aircraft in cruise, which would face their temperatures continually. The Boeing booster would reenter at Mach 7. Its thermal environment, however, would resemble a fire that does not burn your hand when you whisk it through quickly. Across part of the underside, the vehicle would protect itself by the simple method of using metal with more than usual thickness, to cope with the heat. Even these areas would be limited, with the contractors noting that "the material gauges [thicknesses] required for strength exceed the minimum heat sink gauges over the majority of the vehicle."¹¹

In 1954, amid the early feasibility studies that led to the X-15, investigators found, to their pleasant surprise, that they could follow standard aircraft design practice in crafting this research airplane. As a bonus, they also discovered that its metal skin, designed for strength rather than for heat resistance, nevertheless would have enough thickness to serve effectively as a heat sink. Now, in 1971, engineers at Boeing were learning that their booster could offer the same bonus, while providing the convenience of aluminum, the most familiar of metals and the easiest to use.

When McDonnell Douglas went further, introducing an external-tank orbiter that lowered the staging velocity to 6200 ft/sec, its engineers designed a winged booster that was 82 percent aluminum heat sink. Though these

10. Brady and Clauser, *Materials*, pp. 241, 272; Lynch, *Handbook*, pp. 552, 822; *Aviation Week*, January 18, 1971, p. 37; June 7, 1971, pp. 55-61; AIAA Papers 71-804, 71-805; Reports LE 71-7, SD 71-114-1 (North American Rockwell); MDC E0308 (McDonnell Douglas)

11. Report B35-43 RP-11 (Grumman); *Aviation Week*, July 12, 1971, pp. 36-39.

designers could have lowered the staging velocity still further, by putting more propellant in the orbiter's tanks, their selected configuration brought the largest savings in the weight of thermal protection.¹²

Unfortunately, while the move to external tankage brought a welcome reduction in peak annual funding, it took NASA less than one-third of the distance from the \$2.2 billion peak funding of the fully-reusables to the \$1 billion of the OMB. This move had addressed the easiest part of the problem; the rest of the solution would prove considerably more elusive.

The next step came during June 1971, as the new NASA Administrator, James Fletcher, embraced what his agency had previously rejected: a phased approach to shuttle development. This called for the extensive use of interim systems that would make it possible to build and fly an initial shuttle orbiter, with development of the final systems being delayed for several years. Such an approach would be wasteful, for the interim arrangements would cost money and yet would serve merely as a stopgap. Phased development also carried political risks, for in Washington, few things are so permanent as a temporary solution. If NASA could get any kind of shuttle into space, even one of interim design, it might face strong opposition and long delays before it could win permission to build the shuttle it truly wanted. A phased approach, however, would spread the program over a term of years, reducing the all-important peak funding level.

Fletcher had formally taken over his office on May 1. On June 16, in his first major decision, he made an announcement:

The preferred configuration which is emerging from these studies is a two-stage delta wing reusable system in which the orbiter has external tanks that can be jettisoned.

Although our studies to date have mostly been based on a concurrent approach in which development and testing of both the orbiter and booster stages would proceed at the same time, we have been studying in parallel, the idea of sequencing the development, test and verification of critical new technology features of the system. We now believe a "phased approach" is feasible and may offer significant advantages.¹³

12. Report MDC E0376-1 (McDonnell Douglas).

13. Fletcher, statement, June 16, 1971; *Aviation Week*, June 21, 1971, p. 19; June 28, 1971, p. 16; Gomersall and Wilcox, *Working Paper*.

The SSME represented a potentially important element of this approach. It had been slated to receive early attention and funding, for it was to power both Shuttle stages. Its development therefore would pace the entire program. An alternative engine did exist: Rocketdyne's J-2S was an uprated variant of the J-2 engine used in Apollo. It would lack the performance of the SSME, while delivering considerably less thrust. A cluster of these engines nevertheless would push a shuttle orbiter into space, though with a greatly reduced payload weight. Also, use of the J-2S would permit a delay (potentially a long one) in proceeding with the SSME. Indeed, Max Faget was already incorporating the J-2S in a number of his orbiter designs.¹⁴

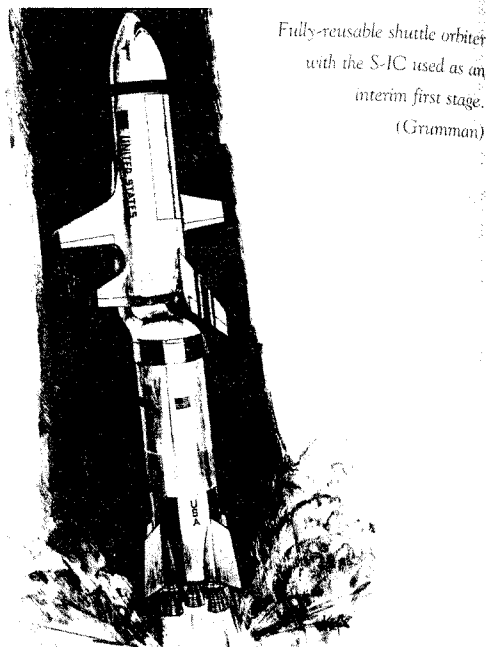
Most of the cuts in peak funding, however, would come by putting off plans to develop a fully-reusable booster, even one of smaller size and simplified design such as one of the new heat-sink versions. NASA instead would fly its orbiter atop an existing rocket stage such as the S-IC, the first stage of the Saturn V. By delaying development of a fully-reusable booster for three years, the agency could cut the peak funding to \$1.3 billion per year. This would represent a reduction of nearly a billion dollars from the proposed peak funding levels of the Phase B studies, and would put NASA within hailing distance of the OMB requirement.

While the move to external orbiter tankage represented ingenuity, Fletcher's decision amounted to desperation. Dale Myers, the Associate Administrator for Manned Space Flight, admitted to *Aviation Week* that phased development might prove to be impractical. Fletcher himself wrote a letter to a leading shuttle critic, the space scientist James Van Allen: "The political cards are so heavily stacked against this program...that no opposition from the scientific community is necessary. I think you are shooting at a dead horse."¹⁵

Nevertheless, the prospect of an interim booster spurred hope among the contractors that had products ready to offer. Boeing's situation now was advantageous, for that company's S-IC was the only immediately-available stage with the power to carry a full-size shuttle orbiter. For Boeing, NASA's new interest in that stage also represented a reversal of fortune. Only a few months earlier, the firm was looking ahead to an imminent shutdown of its

14. Rocketdyne, *Expendable Launch Vehicle Engines*; Jenkins, *Space Shuttle*, pp. 110, 113-115.

15. Letter, Fletcher to Van Allen, July 12, 1971; *Aviation Week*, June 21, 1971, p. 19.



Fully-reusable shuttle orbiter
with the S-IC used as an
interim first stage.
(Grumman)

Saturn production facility. Now it could cherish the hope that production of this rocket stage might continue after all.

Martin Marietta was also in the picture. This firm had actively promoted the Titan III-M, in an era when NASA had been willing to consider a small shuttle orbiter that would amount to an enlarged Dyna-Soar. Though the company had done little of significance for the Shuttle after 1969, now, in mid-1971, it came forward with a new concept, the Titan III-L ("large"). This would use a new liquid-fueled core with a diameter of 16 feet, compared to 10 feet for the standard Titan III, and with up to six solid-fuel boosters of 120-inch diameter.

David LeVine, Martin's vice president for launch vehicles, noted that the solid boosters were already in production at United Technology Corp., and could readily be lengthened to yield more thrust. Though the core would be

new, it too would use existing engines from Aerojet General. Though the Titan III-L would be expendable, it would cost no more than \$30 million per launch, compared to \$73 million for the S-IC. The Shuttle orbiter riding atop the new Titan would certainly not resemble Dyna-Soar; it would be a full-size orbiter, carrying up to 65,000 pounds of payload.¹⁶

The principal builders of solid rocket motors—Thiokol, Aerojet, United Technology, Lockheed Propulsion Co.—were prepared to make a pitch for their own units as well, arguing that suitable clusters could also provide a good interim shuttle booster. A few years earlier, these firms had built and test-fired rockets with diameters of 156 and even 260 inches. This technology had subsequently been set aside as having no immediate application. Now, however, with the Shuttle needing an interim booster, these big solid motors might see their day.

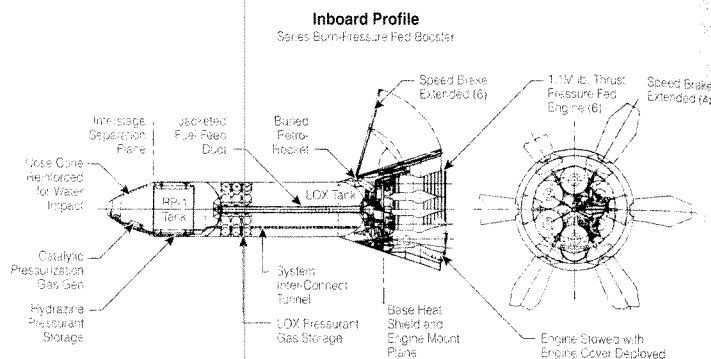
A fourth approach came from NASA Marshall: the pressure-fed expendable booster. It amounted to reinventing the liquid-fueled rocket, using approaches selected for their simplicity. Conventional rocket stages used structures of thin-gauge aluminum that saved weight. Their engines relied on turbopumps that pumped the propellants to high pressures, with the engines operating under internal pressures that were similarly high. Such engines offered good performance and strong thrust from compact and lightweight units. The turbopumps, however, were costly and difficult to develop, and at times were prone to failure.

Marshall's new concept did away with the turbopumps, relying on gas pressure within the propellant tanks to push the fuel and oxidizer into the engines. Though these engines would have to be of new design, their lack of turbopumps would greatly ease the problem of development. They would be larger and heavier than their pump-fed counterparts and would give less performance. Moreover, the booster would gain weight, for it would require heavy thick-walled tanks to contain the pressure. Though this approach was inelegant, flying in the face of the quest for high performance that marked the SSME, it offered one more route toward reducing the peak annual funding.¹⁷

This plethora of possibilities reflected the use of external orbiter tankage, which reduced the staging velocity and made it easier for a booster to do its

16. *Aviation Week*, June 28, 1971, p. 16; July 12, 1971, p. 38; August 2, 1971, pp. 40-41; Report MCR-71-309 (Martin Marietta).

17. Jenkins, *Space Shuttle*, pp. 122-123; *Aviation Week*, June 28, 1971, p. 16.



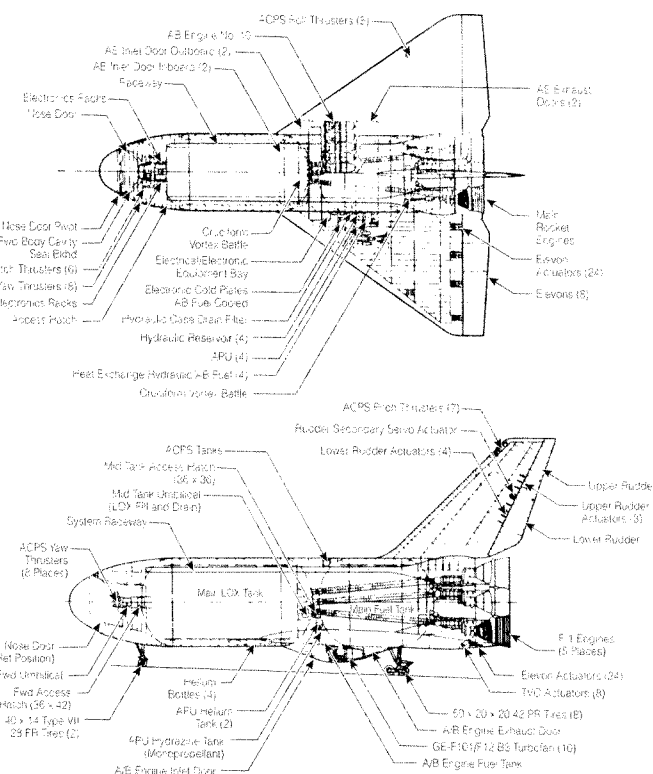
Recoverable pressure-fed booster. (McDonnell Douglas)

job. This diversity of boosters meant that there now was no clear reason to choose any of them. The wide range of alternatives recalled the era of the late 1960s, when a hundred flowers had bloomed and when neither NASA nor the Air Force had yet developed a convincing idea of how a shuttle should look.

Though the contractors might have helped by settling on a preferred type of booster, this happened only in part. Early in September, North American declared that all of them appeared acceptable, and recommended deferring a choice. Grumman also found little reason to prefer any of them—but noted that within an interim program of 30 flights or fewer, the S-IC, built by its partner Boeing, would offer lower costs because it was already in hand and would not demand up-front spending for development. Lockheed gave the nod to a cluster of 156-inch solid boosters. McDonnell Douglas did the same, but also had nice words for the Titan III-L.¹⁸

Having thoroughly muddled the waters with these expendable boosters, NASA officials proceeded to do an about-face as they learned that they might indeed build a reusable booster after all. This happened as further studies of external-tank orbiters showed that they could cut the staging velocity to 5,000

18. Reports SV 71-40 (North American Rockwell); LMSC-A995931 (Lockheed); B35-43 RP-21 (Grumman); Interim Report (McDonnell Douglas).



Winged S-IC stage. (Boeing)

and even below 4,000 ft/sec. This further enhanced the prospects for reducing the booster size and easing the problem of its thermal protection.

Though it lacked wings, the pressure-fed booster became a candidate for reusability because its thick aluminum skin would easily serve as a heat sink. This same robust skin would allow the vehicle to come down by parachute and land in the ocean, surviving the impact and the subsequent perils of the sea. It would enter the water 200 miles offshore; a boat would then bring it back. No

one was prepared to describe this as graceful; with this form of makeshift added to those of its design, the concept was unofficially called the Big Dumb Booster. Such dumbness seemed, however, to promise a new path to wisdom.

A more elegant reusable booster concept came from Boeing, which proposed to remodel its S-IC by turning it into a big airplane. It would receive wings, a tail, a nose with a flight deck, and 10 jet engines for the return to its launch site. Though the standard S-IC had never been built for reentry and reuse, thermal protection would not be a problem; modest thickenings of its aluminum skin now would provide heat sink. To emphasize its near-term feasibility, Boeing's technical artists presented top and side views in lavish detail, even specifying the location of the onboard power units and the choice of tires for the landing gear. At a time when NASA still expected to defer building a flyback booster for several years, one member of the study team emphasized that "our proposal is the reusable booster."

By late September, NASA was ready to abandon phased development, returning to the original plan of simultaneously building the booster and orbiter. The booster would be either the winged S-IC that used standard F-1 engines, or the Big Dumb Booster with pressure-fed propulsion. In turn, the orbiter would use phased technology. It would have a full-size payload bay, but would use four J-2S engines; the SSME program would go on the shelf. For thermal protection, the orbiter would rely on an ablative heat shield, even though this would demand costly replacement after every flight. In time, however, the SSME could be resurrected for use in an upgraded orbiter. This orbiter would also use tiles or hot structures, as reusable and low-cost thermal protection.

The development costs and peak annual funding for the complete Shuttle, in billions of dollars, now looked highly encouraging:¹⁹

	Pressure-Fed Reusable		Winged S-IC	
	Development	Peak	Development	Peak
Grumman	\$ 4.08	\$ 1.02	\$ 4.5	\$ 1.11
North American	5.12	0.94	5.79	1.21
McDonnell Douglas	5.16	0.81	7.51	1.30
Lockheed	(Not given)		4.41	0.99

19. *Aviation Week*, September 20, 1971, pp. 16-17; October 11, 1971, p. 23; October 25, 1971, pp. 12-13; Reports LMSC-A995931 (Lockheed); MDC E-0497 (McDonnell Douglas); B35-43 RP-30 (Grumman); SV 71-50 (North American Rockwell); *Ballistic Recoverable* (Boeing).

These costs compared with estimates prepared by the Aerospace Corp. for a two-stage fully-reusable shuttle: development, \$9.92 billion, with a peak of \$2.34 billion.²⁰ The pressure-fed booster yielded a development cost less than half that value, with a peak comfortably under the OMB ceiling of \$1 billion. The winged S-IC was only slightly more costly, and it preserved all the operational advantages of the fully-reusable boosters of Phase B, on which NASA had placed so much hope.

The new booster concepts did even more, for they addressed major technical deficiencies of the Phase B boosters. Thermal protection had appeared particularly difficult, at a time when tiles were too new to trust. Charles Donlan, the Shuttle program director, put it this way:

Phase B was the first really extensive effort to put together studies related to the completely reusable vehicle. As we went along, it became increasingly evident that there were some problems. You had to develop two hypersonic aircraft configurations, simultaneously. The engine problem was compromised. Trying to use common engines made them not optimized for either case. We ran into a problem of pilot escape from the booster in the event of an abort. We never could quite figure out what to do about it. And then as we looked at the development problems, they became pretty expensive.

We learned also that the metallic heat shield, of which the wings were to be made, was by no means ready for use. The slightest scratch and you are in trouble. It became increasingly evident that you want to have the cheapest possible configuration, but you put all this time and effort on a vehicle, the biggest part of which (the first stage), its only role was to get the orbiter up high enough for it to fly itself. So here you're spending all this effort on a part of the system that had no basic payoff. The important thing is the orbiter—that is the payoff.

So you see, it was easy to say, "Why do we break our necks putting twelve engines in this damn big reusable booster? You cannot get the pilots away, the metallic heat shield is going to give you all kinds of problems, to do it with ablative stuff makes it too heavy; drop it entirely. That started the ball rolling into other types of boosters for the orbiter.

20. Morgenstern and Heiss, *Analysis*, May 31, 1971, p. 0-15.

The merits of an aluminum heat-sink booster appeared stronger yet when one recalls that the most plausible Phase B alternative, which used hot structures in the fashion of the North American Rockwell design, amounted to covering much of the surface of a craft the size of a Boeing 747 with alloys that had previously seen use only in turbine blades. By comparison, the simplicity of aluminum heat sink was overwhelming. It was as much as to say that instead of attempting to leap with one swoop into the metallurgy and launch vehicle design of the twenty-first century, engineers instead could build the booster using design methods that dated to World War II.

The Phase B work had also pointed to demanding issues of safety. To reduce weight to a minimum, the internal tanks had to carry part of the Shuttle's weights and loads. The resulting stresses would tend to produce leaks. Hydrogen leaks are difficult to detect, raising the prospect that hydrogen could build up beneath the skin of an orbiter, to produce a damaging fire or explosion. External tankage solved this problem, at least for the orbiter. A leak of propellant now might indeed be detected, since it would occur in plain view. The leak would also be far less dangerous, for the propellant would waft away on the winds rather than build up to form a dangerous concentration.

The Phase B shuttles also promised severe difficulty in technical development. Airplanes and spacecraft, as a rule, tend to gain weight in the course of development. A two-stage shuttle could accommodate such weight growth by enlarging the tanks, to allow them to carry more propellant. With the tankage being internal in both stages, this would demand extensive redesign of one or both stages—which would introduce opportunities for further weight growth. External tankage addressed this problem as well. A simple lengthening of the tanks would do, leaving both booster and orbiter untouched.²¹

It thus appeared that in mere months, NASA had scored an impressive coup, addressing these technical issues while simultaneously meeting the OMB's cost limits. As with the Mathematica study, however, OMB officials again found reason aplenty to view this work with skepticism.

The new configurations closely suited NASA's institutional arrangements, which dated to the time of Apollo. Those arrangements had given NASA Marshall responsibility for the Saturn launch vehicles and their engines, with

the Manned Spacecraft Center holding responsibility for the Apollo moonship, with its piloted spacecraft and lunar lander. This division of authority had carried over to the Shuttle, with Marshall in charge of the booster and MSC dealing with the orbiter. The Shuttle certainly needed an orbiter; hence MSC's prospects were clear. It was less obvious, however, that the Shuttle needed a booster, particularly if it was to be complex and costly. At Mathematica, Klaus Heiss had managed the Shuttle studies and now was recommending simplified designs within NASA. He met resistance, and as he told the *National Journal*, "For a long time some people over there kept seriously telling us, 'We can't go that route, because we've got to have something for the Marshall Space Center as well as something for the Houston Space Center.'"²²

NASA Marshall had taken shape as a reflected image of Wernher von Braun, its founder and longtime director. He and his fellow veterans of the wartime V-2 effort, who still held senior positions, had long since nailed their flag to the mast of liquid-fuel rocketry, and had left the development of large solid rockets to the Air Force. The recent Shuttle studies reflected this strong Marshall preference. Thus, early in September, the prospect had emerged that the Shuttle might use a cluster of 156-inch solids as its booster. Lockheed and McDonnell Douglas had both made outright selections of this choice, while North American and Grumman had found no reason to reject it.

NASA thus had a clear opportunity to seek a phased-development program with these solids as the interim booster. Instead, in a directive dated September 12, 1971, NASA had instructed its contractors to set aside their studies of phased development and to return to assessments of concurrent development, using two new reusable booster concepts: the pressure-fed and the winged S-IC. Both had NASA Marshall written all over them. Of course, Marshall was the world's leading center for rocket development; its voice would certainly be heard. NASA's decision, however, had amounted to a peremptory dismissal of solids. The results of the new studies were a little too good to be true.²³

The winged S-IC, for one, promised the fully-reusable booster of NASA's hopes. Its shuttle would do nearly everything that the Phase B fully reusables were to accomplish, with a billion dollars less in peak annual funding. Yet

22. *National Journal*, August 12, 1972, p. 1299.

23. NASA Tech. Directive GAC-3, September 16, 1971; Reports B35-43 RP-28 (Grumman), p. 11; SV 71-50 (North American Rockwell), pp. 2, 3, 5.

21. John Mauer interview; Charles Donlan, Washington, October 19, 1983, pp. 19-20; Loftus et al., *Evolution*, pp. 15-18.

THE SPACE SHUTTLE DECISION

because this booster was to grow out of the standard S-IC and would use the F-1 engine, it would continue to demand work aplenty from both Marshall and its contractors. The budget analysts at OMB had been trained to spot flimflam. They had reason for doubt when they saw that in an astonishingly brief time, NASA had succeeded in devising a new shuttle design with a flyback booster that would make the agency's wishes come true—while still purporting to meet the cost goals of the OMB.

Projected cost savings also raised questions, for they seemed disproportionately large when contrasted with the relatively modest engineering changes that had led to the shuttle with the winged S-IC. Were the Phase B cost estimates valid? They were certainly high, and if NASA could receive largesse on such a scale, it could continue to maintain itself in the style to which it had become accustomed. On the other hand, one could equally question the low estimates of the recent exercises. Lockheed's invited particular scrutiny, for it was the lowest—and its peak level, interestingly, was just a hair under the OMB mark of \$1 billion. Everyone knew that Lockheed had won the C-5A contract with an unrealistically low bid; everyone also knew that the man who had presided over that bid, Dan Haughton, was now the firm's chairman. Was Lockheed attempting again to buy in? Were the other contractors very far behind?

Even if one cared to accept everyone's cost estimates as reflecting good faith, the trend of these estimates invited further questions. On its face, this trend meant that in only six months, NASA had cut the planned cost of development in half, from \$10 billion to \$5 billion, while sacrificing little in the Shuttle's capability. These cuts had been won at a price, for the cost per flight was now on the rise—while still remaining low enough, at least in the published estimates, to make the Shuttle attractive to users. Still, it was appropriate to ask if NASA had any other rabbits it might pull from a hat. Thus, late in November two OMB staff members, Daniel Taft and the economist John Sullivan, sent a memo to Donald Rice, an OMB assistant director:

In the light of the innovative Shuttle designs which have been forthcoming over the past several months, we believe that the best procedure would be to provide NASA with a constraint in terms of total investment cost (say, \$3-4 B) rather than have us try to define a preferred configuration. If NASA's resourcefulness to date in changing the Shuttle's design is any guide, we have

A Shuttle to Fit the Budget

*not yet begun to see what they could achieve if they really tried to optimize a system for \$3-4 B.*²⁴

Rice's domain included the Economics, Science, and Technology Programs Division, which covered NASA. He reported directly to the OMB deputy director, Caspar Weinberger. Rice also had noted that NASA's recent exercises in redesign had brought the Shuttle very close to what NASA had wanted originally. Four years later, he noted particularly that NASA's basic assumptions had seemed to be set in concrete:

I guess what sticks in my mind more than anything else about it was the difficulty of getting any solid attention paid to alternative designs. I don't mean alternative in the technical detail sense, but alternative in terms of mission requirements and why that mattered. How hard it was to get an examination of alternative specifications of what you would like to accomplish, and the systems designs that reasonably derived from that will lead to each different specification of what you wanted to do.

He added that within the Pentagon, "they do a hell of a lot better job of looking through the alternatives before they head down one of those roads."²⁵

Even so, there had been change within the Shuttle program, at least in terms of its engineering design. The two-stage fully reusables were dead, and Charles Donlan would not miss them, later declaring, "It wasn't till the phase B's came along and we had a hard look at the reality of what we meant by fully reusable that we shook our heads saying, 'No way you're going to build this thing in this century.' As I say, 'Thank God for all the pressures that were brought to bear to not go that route.'"²⁶

The winged S-IC soon would die as well, for it appeared more costly than the pressure-fed reusable booster. That one might look and fly like an ugly duckling, but it was a graceful swan in the realm of budgets, and would survive into the next round of designs. This round would resurrect the solid-propellant booster, and would determine the shape of the Shuttle in the form that would actually be built.

24. Memo, Taft and Sullivan to Rice, November 29, 1971.

25. John Logsdon interview, Donald Rice, November 13, 1975, p. 1.

26. John Mauer interview, Charles Donlan, Washington, October 19, 1983, pp. 23-24.

End Game in the Shuttle Debate

Within the internal debates of the Executive Branch, the end game, during the second half of 1971, had much in common with the opening gambits of early 1969. Those gambits had featured opposition to NASA's plans from the Budget Bureau, along with the high-level review panel of Charles Townes that had recommended much less than Tom Paine had hoped to pursue. Now, two and a half years later, a similar review would introduce a concept for a mini-shuttle that would win support from NASA's critics—and that this agency could not accept.

Congress was not a significant player within these debates, as it remained generally supportive. The flurry of design changes, in the wake of Phase B, did not dent this support. Congressman Don Fuqua, a leader within the House space committee, notes that these changes initially produced "great consternation" among his colleagues:

We had just finished defending one configuration on the floor and then suddenly they announced they were going to change it. Tiger Teague got the top brass from NASA over here and raked them over the coals.

We all wanted to know how long they had known they were going to change and how much of this kind of thing was going on behind the committee's back. They explained the reasons behind the changes, and everybody calmed down. After that, though events moved pretty fast, they did try to keep us reasonably well informed.²⁷

The President's Science Advisory Committee, however, wanted its own sources of information. Nixon's first science advisor, Lee DuBridge, had retired in August 1970, citing his wish to step down "well in advance of my 70th birthday in 1971." His successor, Edward David, had been executive director of communications research at Bell Labs. David inherited the existing staff of the Office of Science and Technology. Within this staff, Russell Drew soon proposed that a new PSAC panel should review the Shuttle's prospects and offer views on how NASA should proceed. Daniel Taft, an OMB staffer with responsibility for the NASA budget, warmly endorsed this review and urged his management to support it as well.

27. *National Journal*, August 12, 1972, p. 1294.

The resulting panel took shape in mid-1971, with its chairman being Alexander Flax, president of the Institute for Defense Analyses, a Pentagon think tank. In July, at the outset, Rice presented David with a list of questions he hoped that Flax would address. The first meeting of this Flax Committee was a three-day affair in mid-August, at Woods Hole, Massachusetts, far from the heat and humidity of Washington, with Martha's Vineyard visible across the ocean. The group proceeded to meet about once a month, holding discussions with NASA officials and Shuttle contractors. During the summer and autumn, as these meetings proceeded, NASA replayed anew its familiar struggle with the OMB.²⁸

On June 16, when Fletcher announced that he was extending the ongoing contractor studies to consider phased development, he had formally advised the OMB of this in a letter. A month later, in mid-July, Daniel Taft drafted a letter of reply for Don Rice, which asked NASA to "identify an orbiter with minimum performance characteristics." These would include low crossrange, along with substantial reductions in payload size and weight. Rice's letter urged Fletcher to place emphasis on "defining approaches which will substantially reduce the overall investment cost" of the Shuttle, citing a number of design approaches that would cover "the full range of alternatives." Rice, however, did not have to cite a preferred configuration that the OMB would support.

On August 2 the OMB deputy director, Caspar Weinberger, sent a letter to Fletcher that set forth budgetary ceilings for use in preparation of NASA's budget request for FY 1973, which was due at the end of September:

The planning ceilings established for your agency for the 1973 budget are:

(in millions)

<i>Net budget authority</i>	<i>\$ 2,835</i>
<i>Net outlays</i>	<i>\$ 2,975</i>

The above figures provide the basis for developing your 1973 budget submission. The President's budget decisions require that you submit your

28. *Science*, August 28, 1970, pp. 843-844; September 18, 1970, p. 1185; October 23, 1970, pp. 417-419; *National Journal*, August 12, 1972, p. 1295; memo, Taft to Young, January 22, 1971; letters, Rice to David, July 14, 1971; David to Fletcher, July 26, 1971; memo, Drew to PSAC Space Shuttle panel members, August 4, 1971; Myers, Memo for Record, August 19, 1971; NASA, "Documentation of the Space Shuttle Decision Process," February 4, 1972.

THE SPACE SHUTTLE DECISION

budget at or below those figures. [Emphasis in original].... It should also be understood that subsequent developments may necessitate reducing these planning ceiling amounts; thus, there is no assurance that your final budget allowance will remain at this level.

These budget marks, if enacted, would represent a further and substantial cut from recent levels: \$3,268.7 million appropriated in FY 1971, \$3,298 million in FY 1972. With such a budget, NASA certainly would not be able to start the Shuttle during the upcoming fiscal year.²⁹

Weinberger was well aware of this. He also knew that these budget marks reflected staff recommendations that had reached his desk by making their way upward through the OMB chain of command. He was willing to listen to Fletcher's viewpoint as well, and three days later he met with Fletcher and with John Young, the head of OMB's Economics, Science, and Technology Programs Division. In notes from that meeting, Fletcher presented several conclusions:

1. *Come in with budget that meets spec.*
2. *Present several alternatives (incl. shuttle) which bring us back to 3.2 outlay & maybe 3.27 authority.*
3. *Cap...didn't realize manned would be out of business if we made no new starts a la shuttle. Jack Young concurred.*

When Weinberger heard that a \$2.8-billion budget would mean the end of piloted space flight, he suggested that NASA might be able to stay at the FY 1972 level. On August 12, he wrote a memorandum to Nixon:

Present tentative plans call for major reductions or change in NASA, by eliminating the last two Apollo flights (16 and 17), and eliminating or sharply reducing the balance of the Manned Space Program (Skylab and Space Shuttle) and many remaining NASA programs.

I believe this would be a mistake.

1. The real reason for sharp reductions in the NASA budget is that NASA is entirely in the 28% of the budget that is controllable. In short we cut it

29. NASA SP-4012, vol. III, p. 13; letter, Fletcher to Rice, June 16, 1971; memo, Taft to Rice, July 15, 1971; letter, Rice to Fletcher, July 20, 1971; letter, Weinberger to Fletcher, August 2, 1971.

A Shuttle to Fit the Budget



Caspar Weinberger with Nixon. (National Archives)

because it is cuttable, not because it is doing a bad job or an unnecessary one.

2. We are being driven, by the uncontrollable items, to spend more and more on programs that offer no real hope for the future: Model Cities, OEO, Welfare, interest on the National Debt, unemployment compensation, Medicare, etc. Of course, some of these have to be continued, in one form or another, but essentially they are programs, not of our choice, designed to repair mistakes of the past, not of our making.

3. We do need to reduce the budget, in my opinion, but we should not make all our reduction decisions on the basis of what is reducible, rather than on the merits of individual programs.

4. There is real merit to the future of NASA, and its proposed programs. The Space Shuttle and NERVA particularly offer the opportunity, among other things, to secure substantial scientific fall-out for the civilian economy at the same time that large numbers of valuable (and hard-to-employ-elsewhere) scientists and engineers are kept at work on projects that increase our knowl-

THE SPACE SHUTTLE DECISION

edge of space, our ability to develop for lower cost space exploration, travel, and to secure, through NERVA, twice the existing propulsion efficiency for our rockets.

He warned against canceling Apollo 16 and 17, noting that such action...

...would have a very bad effect, coming so soon after Apollo 15's triumph. It would be confirming, in some respects, a belief that I fear is gaining credence at home and abroad: That our best years are behind us, that we are turning inward, reducing our defense commitments, and voluntarily starting to give up our super-power status, and our desire to maintain our world superiority.

America should be able to afford something besides increased welfare, programs to repair our cities, or Appalachian relief and the like....

7. I believe I can find enough reductions in other programs to pay for continuing NASA at generally the \$3.3 - \$3.4 billion level I propose here. This figure is about \$400 - \$500 million more than the present planning targets.

Here was a milestone. For the first time since the heyday of Apollo, NASA now had an advocate who had real clout within the budget and policymaking process. What was more, Weinberger's memo brought a response, as Nixon read it and wrote in the margin. "I agree with Cap."³⁰

By mid-August George Low, the deputy NASA administrator, had reason to believe that piloted flight might survive, if only at a bare-bones level. He wrote, "My own view is that we might be able to bring the 1973 budget back to the 1972 level, but that our chances of bringing it above that level are essentially nonexistent." The 1972 budget had been too small to launch the Shuttle as a new start, and if future budgets would continue at that level, a shuttle would be out of the question. Taking the bull by the horns, he added that in this situation, "we should drop the Shuttle right now":

My view is that we should assume that this is a permanent situation, that we should drop the shuttle, and that we come up with a new manned space flight program. In my view, this program should be based on an evolutionary space

30. Fletcher, meeting notes, August 5, 1971; Weinberger, Memorandum for the President, August 12, 1971. Reprinted in part in NASA SP-4407, volume I, pp. 546-547. For Nixon's comment see attachment to memo, Huntsman to Shultz, September 13, 1971.

A Shuttle to Fit the Budget

station development, leading from Skylab through a series of research and applications modules to a distant goal of a permanent space station.... The transportation system for this manned space flight program would consist of Apollo hardware for Skylab; a glider launched on an expendable booster for the research and applications modules; and finally, the shuttle but delayed 5 to 10 years beyond our present thinking. The new element in this plan is the expendable booster-launched glider. The whole program ties together in that none of it is dead-ended. The glider would be both an up and down logistics system for the research and applications modules, and, at the same time, lead toward the development of a shuttle in the future.

The glider would lack engines; it thus would resemble, once again, an enlarged Dyna-Soar. Martin Marietta had advocated such vehicles as recently as 1969; now, with its proposed Titan III-L, it had a launch vehicle suited to this task. Low's notes amounted to admitting that with piloted flight *in extremis*, this glider might form the basis of a fallback position that NASA might accept if continued budget reductions were to force the agency to abandon its plans for a Shuttle.

Two days later, in a review of Shuttle configurations and alternatives, Low described the glider more fully:

The glider itself would look somewhat like the shuttle, but would be smaller. It would carry a payload of 12 feet by 40 feet and a payload weight of about 30,000 pounds. It would have sufficient propulsion for on-orbit maneuvering but would not have the engines or propellant tanks required to propel itself into orbit. It would make use of current technology in avionics and other on-board systems. The glider would be placed into orbit with a two-stage vehicle of the Titan III-L class. The glider and its payload would be reusable but both booster stages would be expended. The requirement for a 15 ft. by 60 ft., 65,000-lb. payload, could be met with the same expendable launch vehicle.³¹

Needless to say, this was not an approach that NASA would embrace willingly. The agency was still a gung ho outfit, deeply engaged even at that

31. Low, Personal Notes No. 52, August 15, 1971; Low, "Space Transportation System Planning," August 17, 1971.

moment in sending astronauts to the Moon, and its senior officials were not about to go back to Dyna-Soar if they could avoid it. Dale Myers, head of the Office of Manned Space Flight, wrote a memo to Low in which he scoffed at the glider, declaring that its advocates wanted no more than to send an astronaut "whirling about the Earth" with no evident use for him in space. In the meantime, these people can get back to doing things as they do them now, with various sized ballistic systems, a relatively constant budget, and a relatively 'status quo' sort of operation."

LeRoy Day, deputy director of the Shuttle program, would criticize the glider more pointedly:

You had to put this thing on top of an enormous booster which you had to throw away each time. And so you had an operating cost that was getting to be kind of ridiculous. The vehicle size and everything—it didn't have much utility. It was kind of a demonstration. It would certainly have been a research vehicle that you could have studied re-entry with. When you got all through with that then you would have said, "Gee, that would be nice if it was big enough to really do something." Then you would have to turn around and build another vehicle. And with the way the budget climate looked, we were pretty sure that we'd be shut out. We'd never be able to say, "Okay, now let's start up a real program and build another one that will be an enlarged version and have more capability." The OMB and Congress would never support it; it would be like two different programs, and we said, "That'll be the death of it for sure."³²

Then in mid-September, just when it counted, George Shultz, director of the OMB, received a staff memo:

The President read with interest and agreed with Mr. Weinberger's memorandum of August 12, 1971 on the subject of the future of NASA.

Further, the President approved Mr. Weinberger's plan to find enough reductions in other programs to pay for continuing NASA at generally the 3.3–3.4 billion dollar level, or about 400 to 500 million more than the present planning targets.

32. Memo, Myers to Low, September 29, 1971; John Maher interview, Leroy Day, October 17, 1983, p. 29.

Significantly, this approval did not embrace the Space Shuttle itself. Low writes that during the next two weeks, "Fletcher and I debated whether we should not forego the Shuttle entirely and develop instead some alternative manned space flight program." Late in September, however, in presenting NASA's formal budget request for FY 1973, Fletcher screwed his courage to the sticking point and indeed sought funds for a Shuttle.

This budget request took the form of a 13-page letter to Shultz. Fletcher stated, "My minimum recommended program requires budgetary authority of \$3,385 million and outlays of \$3,225 million for FY 1973." He placed the Space Shuttle at the top of his agenda, and asked for funding of \$228 million: \$200 million for research and development, and \$28 million for facilities. Three weeks later, in a follow-up letter to Weinberger, he emphasized the Shuttle anew:

Space shuttle development should be started no later than the summer of 1972 [emphasis in original]. Our studies have led to a shuttle concept that minimizes peak funding requirements, and at the same time drastically reduces development cost required for the first manned orbital flight. Yet this concept still leads to a productive space transportation system to meet the needs of U.S. civilian and military space programs.³³

Fletcher, however, was not the only one with ideas on how to build a piloted spacecraft. Within the Flax Committee, reviewing the Shuttle program on behalf of the White House, a member named Eugene Fubini had taken a leading role. He had been the Pentagon's deputy director of research and engineering, and he had willingly met with the same officials of Martin Marietta who had been recommending the Titan III-L and its glider in lieu of a true Space Shuttle.

Influenced by Fubini, Flax sent an interim report to Edward David on October 19. This report presented a set of alternatives in which the glider, far from ranking lower than the least acceptable form of piloted space vehicle, was actually the most ambitious option that the Flax Committee was willing to endorse. Moreover, the committee's glider would carry only 10,000 pounds

33. Memo, Huntsman to Shultz, September 13, 1971; letter, Fletcher to Shultz, September 30, 1971; Low, Personal Notes No. 55, October 2, 1971; letter, Fletcher to Weinberger, October 19, 1971.

of payload, one-third as much as the version hesitantly considered by NASA's George Low.

What could be less ambitious than this glider? There was the possibility of modifying Apollo spacecraft to make them refurbishable and continuing to fly them using the existing Saturn I-B, on rare occasions. Another option called for developing the Titan III-M and its Big Gemini, a variant of this existing spacecraft that would grow to carry as many as nine people. Such alternatives might keep piloted space flight alive—but it would resemble a patient on life support.

Flax's report also addressed the subject of shuttle economics. He dealt specifically with NASA's preferred concept, which called for concurrent development of booster and orbiter, with the orbiter using phased technology. People described this orbiter as "Mark I/Mark II," referring to an initial version that would later be upgraded with SSME engines, reusable thermal protection, and advanced onboard electronics.

NASA's estimated cost per flight was \$9.0 million for Mark I, falling to \$5.5 million for the improved Mark II. Significantly, and like the OMB, Flax did not challenge these estimates. He merely denied that they promised advantage. As his report stated,

Considering all of the technological and operational unknowns involved in the shuttle development and the fact that no vehicles of similar function have ever been designed before or have ever operated over the range of flight regimes required for the shuttle, prudent extrapolation of prior experience would indicate that estimated development costs may be 30 to 50 percent on the low side. Thus, the estimates of \$6.5 billion in RDT&E³⁴ for the Mk I/Mk II shuttle program may range between \$8.5 to \$10 billion, reflecting increased program costs of \$2.5 to \$3.5 billion. Similar uncertainties must be considered to apply to other non-recurring costs such as production and facilities (amounting to about \$4 billion). Thus a possible cost uncertainty of about \$5 billion for total program costs might be envisioned giving a high estimate of total non-recurring cost of about \$15 billion.

At a launch rate of about 40 per year (DOD, NASA and other) over the 13 years used in the NASA cost model and an average payload cost of \$30

million (not unrepresentative of the mix of current unmanned payloads), the total payload costs would be \$15.5 billion. Thus, even if the total payload cost were saved (including those launched to Mars, Venus, etc.) over a 13-year period by recovery and reuse at zero refurbishment cost, it would, in the case of the high-end cost estimate, barely offset the cost of the shuttle program without discounting. A more realistic (although probably generous) estimate of the savings possible through payload recovery might be 50 percent of payload costs which could account for only \$7.5 billion.

The other area of savings which is offered by the shuttle is its launch cost. Average launch cost with current expendable boosters is \$12 million (projected into the 1978-90 era in the NASA cost model). Thus, with current expendable boosters, the annual launch cost will be \$500 million. The cost of Mk II shuttle operation per flight is usually cited at \$5.5 million; thus the cost for 40 flights per year will be \$220 million. The saving of \$280 million annually for 13 years amounts to \$3.6 billion. However, a doubling of the operational cost would reduce the saving to \$60 million annually or \$780 million....

The operating cost estimates of \$5.5 million per flight for the shuttle, within narrow limits, must be considered to be a very rough estimate at this time, particularly for the early years of shuttle operation. The actual value will depend upon the time between overhaul of equipment not yet designed, refurbishability of thermal protection system materials not yet out of the laboratory, and on the feasibility of operating in the shuttle in an "airline" mode radically different from all past experience in space operations.³⁵

These few paragraphs delivered a body blow to the Shuttle's economic prospects, for although their conclusions were highly unfavorable, they actually carried a strong bias in favor of the Shuttle. This was because this analysis used current or undiscounted dollars. The Mathematica study had moved heaven and earth to try to justify the Shuttle in dollars discounted at ten percent per year, and members of the Flax Committee had little use for that study's findings. In the words of a committee staff member, "No one believed all the fancy economics and no one believed the mission model. I think they were on

34. Cost of development: Research, Development, Test and Engineering.

35. Note, Frank Williams to Von Braun, August 23, 1971; Low, Personal Notes No. 56, October 17, 1971; letter, Flax to David, October 19, 1971.

THE SPACE SHUTTLE DECISION

hemph when they were talking about sixty flights per year." Yet it had seemed easy to show that the Shuttle could be cost-effective in undiscounted money.

Now, however, Flax was saying that the Space Shuttle would fail this test as well. NASA might have moved mountains to try to cut the Shuttle's cost of development, but the complete nonrecurring cost, taking account of a plausible overrun, could easily approach \$15 billion. The undiscounted savings, with a reasonable mission model and an optimistic cost per flight, would barely top \$11 billion and could be less. The OMB economist John Sullivan, who had made shuttle studies his specialty, summarized the matter within a memo to Don Rice: "A Shuttle cannot be justified when using cost-effectiveness as the criterion."

Nevertheless, no one wanted to shut down the piloted-flight effort. In early October, Low wrote of a meeting with the OMB's John Young:

I took Jack Young to lunch about a week ago, largely because I had heard that he was the one most negative toward manned space flight within OMB. During our discussions, he agreed that the manned space program and the unmanned planetary programs were the big swing factors in the NASA budget. He indicated, however, that he understood that only the level of manned space flight was in question and not whether to have a manned space flight program at all. He agreed that the President would not, and could not, stop the nation's manned space flight effort.³⁶

Yet if piloted space flight was to involve more than infrequent missions in which astronauts would show the flag, then NASA would need help, and in a hurry. The help came from Mathematica, which now gave a strong endorsement to a preferred shuttle configuration.

TAOS: A New Alternative

The Mathematica analysis of May 31, 1971, had drawn criticism because it had dealt only with the two-stage fully-reusable approach of Phase B, paying no heed to design alternatives. Yet a comparison of such alternatives, on eco-

36. Memo, Sullivan to Rice, October 19, 1971; Low, Personal Notes No. 55, October 2, 1971; John Logsdon interview, Dave Elliot, p. 3.

A Shuttle to Fit the Budget

nomic grounds, promised insights that even the Flax Committee was willing to pursue. Flax's interim report stated that while direct cost-benefit analysis lacked "sufficient credibility to serve as a primary basis for deciding to undertake such an expensive and high-risk program," economic comparisons "are undoubtedly extremely valuable in making cost tradeoffs and in considering alternatives in design and program planning."

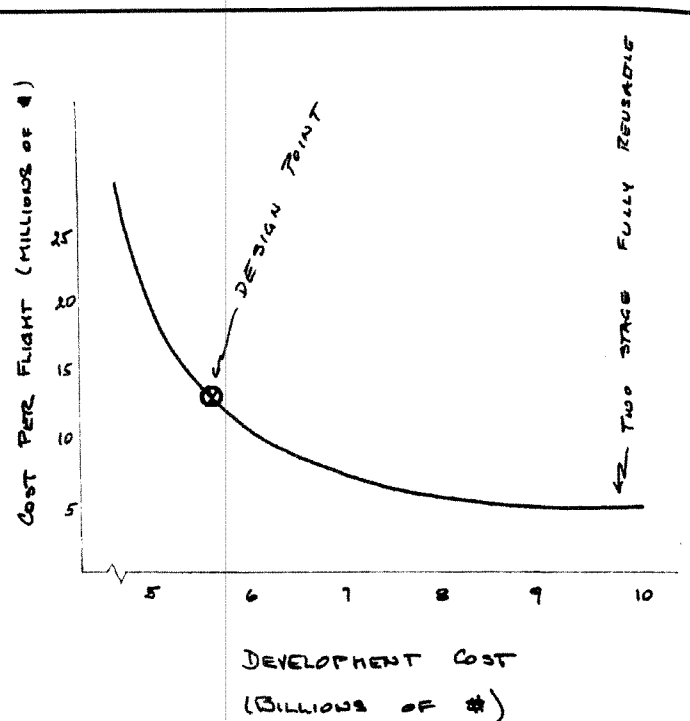
In studying such alternatives, designers traded reduction in development cost against an increase in cost per flight. This tradeoff had to stay within bounds; if the cost per flight was too high, the Shuttle would fail to capture traffic from existing expendables. Yet the Mark I orbiter, as discussed in Flax's report, would cost \$9 million per flight, comfortably under the \$12 million of expendables. This was enough of a margin to give this design a strong advantage.

Klaus Heiss, who was continuing to direct the Mathematica studies, was not in a position to introduce new design concepts of his own. Instead he proceeded by receiving the contractors' concepts, comparing them using methods of economics. As he pursued such comparisons, he concluded that NASA had not pushed its tradeoffs far enough. The opportunity existed to push the cost per flight as high as \$10 million, in exchange for a further reduction in development cost and in peak annual funding. Even at \$10 million per flight, all but five percent of the Shuttle's planned missions would remain cost-effective when compared with similar missions using expendable launch vehicles.

Heiss saw that a specific class of Shuttle designs would do this. He called this class TAOS (Thrust Assisted Orbiter Shuttle). It would use a standard orbiter, possibly of the Mark I/Mark II type, with an external tank large enough to permit the orbiter's engines to operate with "parallel staging," burning from liftoff to orbit. Two booster rockets would flank this tank, giving added thrust after liftoff, then falling away at staging velocity. The boosters might use pressure-fed liquid-fuel engines. Alternately, they could use solid propellant.

"If you could go to \$10 million," Heiss later told *National Journal*, "then some kind of thrust-assisted orbiter shuttle beat out all other systems. It had the lowest development costs of any system capable of sustaining continuous manned flight."³⁷

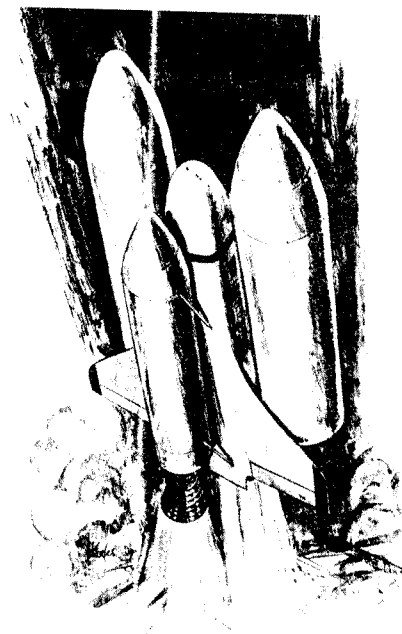
37. Letter, Flax to David, October 19, 1971, p. 6; *National Journal*, August 12, 1972, pp. 1298-1299.



Sketch made by George Low in 1979, illustrating the trade-off between cost per flight and development cost. (Drawing by George Low courtesy of John Logsdon)

Such designs already existed. McDonnell Douglas had one called RATO, Rocket Assisted Take-Off. It put the orbiter's propellants in two external tanks, with a 180-inch solid rocket mounted to the belly. Grumman had TAHO, Thrust Assisted Hydrogen-Oxygen, with a single external tank flanked by twin pressure-fed liquid boosters. This background meant that key NASA contractors already had people who had introduced TAOS concepts and had studied some of their engineering issues.

For TAOS, the 260-inch solid motor would not do. It would be difficult to develop, hard to handle due to its enormous size, and yet it would offer no



Example of TAOS. Grumman concept of a shuttle with a single large solid-propellant booster and propellants in external tankage. McDonnell Douglas had a similar concept called RATO (Rocket Assisted Take-Off). (Grumman)

cost advantage. The TAOS approach, however, specifically included the selected Shuttle configuration of 1972, with twin solids, smaller in size, for the booster. Ironically, this specific configuration appears to have originated within the Institute for Defense Analyses, the think tank whose president, Alexander Flax, was challenging NASA's approaches so effectively.

The IDA had close ties to the Air Force, which had been using solid boosters for years on the Titan III, and NASA's Charles Donlan gives credit for the twin-solid TAOS to two IDA staffers, Reinald Finke and George Brady. In 1986, Donlan declared that "Brady came up with the configuration that's almost identical" to the one that NASA built. "And so a year or so ago,

I recommended that NASA give him one of their awards, when they were recognizing Shuttle contributors."

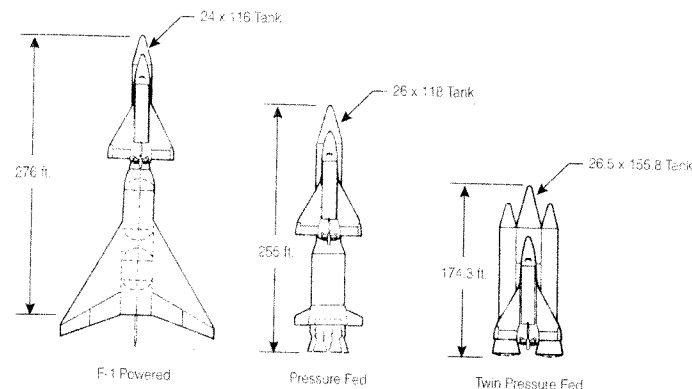
During 1971, however, TAOS had a strong disadvantage: those booster rockets, whether with liquid or solid propellant, would be unpiloted. NASA still was pinning its hopes on a piloted booster, quite possibly the winged S-IC, and its officials paid little heed to Heiss's proposals. Heiss knew why: "In the first place, there was the irresistible urge to go for the most advanced design and technology possible. And then there was a deep-seated bureaucratic bias for two manned vehicles." There also were the institutional prerogatives of NASA Marshall, which might accept pressure-fed liquid boosters in a pinch, but had little background in solids. The OMB's Don Rice would note that NASA "pushed so hard for the liquid fuel thing in the first place, because it was hard to find something for Marshall to do. A reason to keep Marshall around if they didn't have one of those big booster development programs underway."³⁸

Heiss nevertheless pressed the case for TAOS within NASA, working with Robert Lindley, the director of engineering and operations within the Office of Manned Space Flight. Lindley had carried out initial cost-benefit studies of the Shuttle, as an in-house NASA effort, and had worked closely with Heiss when Mathematica pursued its subsequent efforts. Heiss would describe Lindley as "one of the few people over there to grasp clearly the real economic and design tradeoffs."

Through September and October, Heiss and Lindley exhorted McDonnell Douglas and Grumman to include TAOS designs in their presentations to NASA. Says Heiss, however, "some NASA officials kept telling them to forget it, the configuration had no chance." As late as October 15, in a presentation to the Flax Committee, NASA failed to include TAOS. Toward the end of the month, Heiss adds, "we thought the whole program was on a catastrophic course."

"There were still many people in NASA who believed they could sell the Administration an \$8-billion to \$10-billion two-stage flyback system," which might be even more ambitious than the phased-technology orbiter with winged S-IC. "At the other end of the spectrum, the OST [White House Office of

38. Report B35-43 RP-28 (Grumman); *Interim Report* (McDonnell Douglas), pp. 13, 25; John Mauer interview, Charles Donlan, October 19, 1983, pp. 28-29; John Logsdon interview, Donald Rice, November 13, 1975, p. 2; *National Journal*, August 12, 1972, p. 1299.



Booster concepts of autumn 1971, with a winged S-IC at left. Configuration at right is a TAOS. All three alternatives were to boost the same orbiter, which would carry all its propellant in a single expendable tank. (McDonnell Douglas)

Science and Technology] and OMB seemed to be drawing back toward some kind of advanced expendable system," based on the Titan III-L or the even less capable Titan III-M. A major shuttle design review loomed during November, and in Heiss's words, "We decided to try and get to Fletcher."

Heiss and his boss, Oskar Morgenstern, wrote a 15-page report that called strongly for TAOS. Under standard procedures, they would have submitted it through channels; senior officials with axes to grind could have sat on it or downplayed it before the report could reach the policymaking level. Instead, Heiss and Morgenstern sent it to Fletcher directly. "Those who say Mathematica was a kept child of NASA really don't know what they're talking about," Donlan later said. "Morgenstern was nobody's kept child. In fact, we had no way of controlling what he did."³⁹

The conclusions of his report dealt not only with the Shuttle but addressed the Space Tug, a reusable rocket stage intended to carry spacecraft to and from high orbit:

39. *National Journal*, August 12, 1972, p. 1299; John Mauer interview, Charles Donlan, October 19, 1983, p. 28.

The TAOS concept foregoes the development of a Two Stage Shuttle System. With the use of thrust assists of either solid rocket motors or high pressure feed systems—which can be made in part reusable for low staging velocities—the TAOS concepts promise a reduction of the non-recurring costs (RDT&E and initial fleet investment) from about \$9 billion or more (two stage systems, including reusable S-IC) to about \$6 billion or less, with a minimal operating cost increase, if any, in the operating phase of the TAOS system.

The detailed economic justifications of the TAOS concept—when compared to any two stage reusable system are:

1. The non-recurring costs of TAOS are estimated by industry to be \$6 billion or less over the period to 1979 or to 1984-85, depending on the objectives and choices of NASA.
2. The risks in the TAOS development are in balance lower but still substantial. Inert abort with external hydrogen and oxygen tanks is feasible; lagging performance in the engine area can be made up by added external tank capacity. A large reusable manned booster is not needed.
3. The TAOS's that were analyzed promise the **same capabilities** as the original two stage shuttle, including a 40,000 pound lift capability into polar orbit and a 60 x 15 feet payload bay.
4. The TAOS can carry the **Space Tug** and capture high energy missions from 1979 on.
5. The most economic TAOS would use the **advanced orbiter engines** immediately. Our calculations indicate that among the alternative TAOS configurations an early full operational capability (i.e., high performance engines on the orbiter) is economically most advantageous, and feasible, within budget constraints of \$1 billion peak funding.
6. The TAOS **can** use J-2S engines on the orbiter for an interim period.
7. The TAOS **abolishes** completely the immediate need to decide on a **reusable booster** and allows postponement of that decision without blocking later transition to that system if still desired. Thereby, TAOS eliminates or lowers the risk and potential cost overruns in booster development.
8. The TAOS can use "**parallel burn**" concepts, which, if feasible, may change the reusable booster decision.

9. Technological progress may make **tank costs**, and thrust assisted rocket costs less expensive, thus further aiding TAOS concepts when compared to two stage concepts.
10. TAOS assures NASA an **early program definition**, and a purpose to the agency. An agreement on TAOS will allow NASA Headquarters a quick and clear reorganization of major NASA centers to meet the TAOS development requirements economically.
11. The TAOS funding schedule makes an early Space Tug development possible. The Space Tug is an important part of the Space Shuttle System. A 1979 Space Tug should recover its complete development costs before 1985 even with the stretched build up of Space Shuttle missions from 1979 to 1985.
12. A clear policy on TAOS development will give an incentive to European countries to undertake and fund the Space Tug development—thereby possibly even eliminating Space Tug funding from NASA budget considerations.
13. The cost per launch of TAOS can be as low as \$6 million or even less on an **incremental cost** basis, with reuse of parts of the thrust assist rockets (either SRM or pressure-fed). With Point 9 realized, the costs of TAOS would practically match the **costs per launch of the two stage fully reusable system**.
14. TAOS practically assures NASA of a reusable space transportation system with **major objectives achieved**.⁴⁰

Like Grumman a few months earlier, Mathematica, with Point 13, was not above assuring Fletcher that he might have his cake and eat it too. This low cost per flight indeed might be attained—but only if it proved possible to dunk a pressure-fed booster into the Atlantic and fish it out none the worse for wear, ready for refurbishment and reuse at minimal cost. Contractor studies, however, soon supported Mathematica's high hopes, showing that TAOS indeed offered low cost—and that solid boosters promised costs that were lower still:⁴¹

40. Heiss and Morgenstern, *Factors*. Reprinted in NASA SP-4407, vol. I, pp. 549-555.

41. Reports MDC E0497 (McDonnell Douglas), p. 1-23; B35-43 RP-30 (Grumman), pp. 2, 17.

THE SPACE SHUTTLE DECISION

Concept	Development cost \$ B	Peak annual funding \$ B	Cost per flight \$ M
Grumman: twin pressure-fed	4.02	0.97	8.2
McDonnell Douglas: twin pressure-fed	4.83	0.74	6.4
McDonnell Douglas: twin solid motors	4.34	0.71	9.9

TAOS did not take NASA by storm. Its design concepts were already part of the mix of alternatives; they had merely been languishing for want of attention. TAOS, however, proved highly useful as it provided a fallback concept, less costly than the winged S-IC, which NASA would adopt as it continued to yield to pressure from the OMB.

A Time to Decide

For NASA, it was time to have the OMB fish or cut bait. Fletcher had submitted his budget request on September 30, with a line item of \$228 million to initiate Shuttle development. This NASA request was one of many budget proposals from federal agencies and cabinet departments; the OMB would modify them and then assemble them into the President's budget, which would go to Congress early in 1972. An early milestone in this budget process, the Director's Review, included a session titled "Space and General Research." Weinberger ran those reviews in 1971. They represented meetings at which OMB staff would recommend actions on agencies' requests, with Weinberger being free to respond. Following standard rules, NASA officials would not be present.

OMB staff members were well prepared for this session. A staff paper, prepared in early October, admitted that the use of a single pressure-fed booster would be more cost-effective than use of the winged S-IC. This represented a response to NASA's booster alternatives of September; the pressure-fed option was a rocket stage of the usual type, and the Shuttle was not a TAOS. While this configuration looked better than the alternative, it still was not good enough. The paper concluded that the most cost-effective option of all, more so than any shuttle, was an uprated Titan III with Big Gemini.

A Shuttle to Fit the Budget

By contrast, Fletcher wanted his shuttle and he wanted it very soon. In a letter to Weinberger of October 19, he pressed his point:

The aerospace industry will be hurt by continuing indecision and further delay in the shuttle program. A firm go-ahead, on the other hand, will quickly create jobs in the industry.

It will not be possible to sustain the momentum now built up in the shuttle program much longer. A loss in momentum will have serious and costly consequences, and may even be irreversible.⁴²

It would not be possible to maintain the Shuttle effort indefinitely as an exercise in design study and analysis. The aerospace industry was set up to build, not to dither endlessly. Its preliminary-design groups, which had carried forward the Shuttle work to date, practiced a highly skilled and specialized trade. If their efforts were not to bear fruit, their companies would be weakened unless senior management could reassign these people to the preparation of other proposals.

NASA's Director's Review session took place on October 22, and the OMB staff recommendation was blunt: cancel the Shuttle program. Staff members also proposed that if this was not feasible, the decision should be held off for another year, when the OMB would deal with the budget for FY 1974. Weinberger was not eager to accept such options. Neither was he willing to flatly override his staff, even though this was his prerogative, and endorse NASA's position. NASA wanted the Mark I/Mark II orbiter with winged S-IC booster, and Weinberger was interested in alternatives studied by the Flax Committee.

That committee had held a key meeting only a week earlier, with the OMB staff economist John Sullivan attending as an observer, and Flax had reviewed shuttle alternatives only three days earlier, in his interim report to the White House's Edward David. These alternatives included the glider. As Weinberger later described the meeting:

The staff, in effect, decided they weren't going to do it. The staff then told me that, if we wanted to do it, it could be done less expensively. So I was

42. Letters: Fletcher to Shultz, September 30, 1971; Fletcher to Weinberger, October 19, 1971; NASA, "Documentation of the Space Shuttle Decision," February 4, 1972; *Space Policy*, May 1986, p. 112.

delighted to hear that; and so they went back and worked with NASA to work out a different configuration.

I could have cut it off at the Director's Review and insist that we are going to do it the way NASA wants it. But the opportunity to do it at a lower cost on additional analysis appealed to me. I never had any doubt in my own mind but, one way or another, I wanted to do it. I thought it was the proper thing for the government to do at that time, and that we needed some forward-looking new activities geared toward the future instead of the past.

With Solomonic precision, Weinberger then split the difference between the views of NASA and of his staff. He told Don Rice that some sort of shuttle indeed would be approved; NASA would not have to settle for Big Gemini. Rice and his staff, however, would have a free hand to seek lower costs. If this meant that NASA would wind up with nothing more than the glider—well, no one in Washington expected the OMB to come forward as a bearer of gifts.⁴³

NASA had some support on the White House staff, but its critics were both numerous and well-placed. In mid-November, George Low gave a summary view of the players that reflected the assessments of William Anders, a former astronaut who now was running the interagency National Aeronautics and Space Council:

Weinberger: is a real space buff. The only one in OMB really positive toward the NASA program. Causes Rice to over-balance in the opposite direction. Everybody lower in OMB is negative.

Rice: the most knowledgeable opposition comes from Rice. Feels that NASA is out of control; however, he will probably support a glider on a TITAN III.

Ed David: was noticeably quiet, measuring his words, and repeatedly saying that he only represented science and that other factors are also involved.... Not really plugged into the President.

Flax: Fubini is really running the Flax Committee. Flax apparently states that no program as large as the Shuttle will gain continuing support. We need a less costly program.... Anders feels that Flax is driving David toward the glider and not vice versa. Anders believes David will support the Orbiter

43. NASA, "Documentation of the Space Shuttle Decision," February 4, 1972; John Logsdon interview, Caspar Weinberger, San Francisco, August 23, 1977, pp. 5, 13; *Space Policy*, May 1986, p. 113; *Science*, May 30, 1986, pp. 1102-1103.

with the parallel staged pressure-fed booster if Flax so recommends.

Whitehead: Whitehead could be helpful in making Flanigan a meaningful communications link to the President, which Anders believes Flanigan needs to be. Whitehead's main motivation now is to improve the Fletcher/Flanigan communications link. Whitehead can be extremely helpful in selling the NASA desired Shuttle approach.... Believes in a \$3.5 billion NASA.

Rose: is the California unemployment buff in the White House. Tries to be helpful and sees Flanigan all the time. He defers to Whitehead when Whitehead is present.

Flanigan: states that the Shuttle story is improving; however, he is by no means convinced that there should be a Shuttle. Is strongly influenced by Whitehead, Rose and David.

Peterson: is the most negative of all about NASA. Perhaps the most dangerous opposition we have within the White House. Believes the space program is the place to take money to stimulate technology. Asked why not take \$1 billion out of space and who needs manned space flight.

Ehrlichman: asked the question, "Given the public attitude on space, why not put the money in aeronautics?" However, he is very much concerned about the aerospace industry and will probably go along with whatever OMB/OST/Flanigan recommend.⁴⁴

These were not the scoundrels depicted in the subsequent bestseller on Watergate, *All the President's Men*. Though skeptical, they were open to argument and perhaps even to persuasion, and it was up to Fletcher and Low to try to sway them.

On November 22, Fletcher responded to a request from Jonathan Rose, a member of Flanigan's staff, and sent him a 10-page paper that presented NASA's case for the Shuttle. On that same day, Low addressed a request from Rice and offered his own paper, which gave a reply to Flax's interim report of a month earlier. Flax had discussed a range of alternatives that included Big Gemini and a small glider. Low now gave his own range of alternatives, with each receiving a succinct description. Big Gemini was conspicuous by its absence, but in other respects, Low's list covered the most important con-

44. Low, Memo for Record, November 15, 1971.

figurations of 1971.

At the high-cost end of this list were two-stage reusables, both the fully-reusables of Phase B and the variant with external hydrogen tanks on the orbiter. In the middle was the Mark I/Mark II orbiter with a single large tank carrying both propellants, and four choices of booster: winged S-IC, single pressure-fed first stage, TAOS with twin pressure-fed boosters, TAOS with twin solids. Anchoring the low-cost end was the Titan III-L with a glider carrying payload of 30,000 pounds. Low recommended a version of TAOS: "The most promising candidate configuration today is the Mark I/Mark II orbiter with the parallel-staged pressure-fed booster," with all engines burning at liftoff.

Low also went to a meeting of the Flax Committee and drew a diagram on a blackboard, summarizing the points of this paper. Fletcher sent a copy to Edward David at the White House, with a cover letter:

All of these configurations of the Shuttle can be developed for costs substantially below those we planned six months ago. We have progressed to the point where a decision to proceed with the shuttle in connection with the FY 1973 budget process is definitely in order.⁴⁵

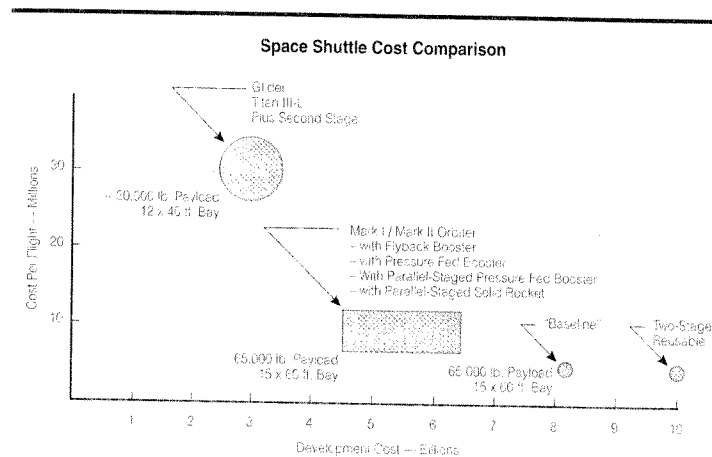
Needless to say, the OMB staff had its own views of the matter. Low's glider had a 12 x 40-foot payload bay; it was the lowest in his range of options, but was the most ambitious of OMB's. On November 29, OMB staffers Daniel Taft and John Sullivan sent their own memo to Rice:

Even a small (10' x 20') glider could capture all manned space flight traffic, e.g.: Space station visits and return to Earth (station modules could be launched unmanned by expendable rockets).

Since the Shuttle is not an economic system under optimistic assumptions, the importance of whether or not all of the payload benefits are realized becomes less significant. The important factors are really such considerations as national prestige, continuation of a manned space flight program, and advancement of technology. Any of the reusable vehicles discussed in this paper (even a 10' by 20' glider) provide this type of intangible benefit.

A 10' x 20' glider would provide virtually the same intangible benefits as

45. Memo, Fletcher to Rose, November 22, 1971; letters: Low to Rice, November 22, 1971; Fletcher to David, November 24, 1971; Low, Personal Notes No. 59, November 28, 1971.



Alternate concepts of November 1971, showing cost per flight and development cost. Compare with figure on page 374. (NASA)

a 15' x 60' orbiter for less than half the investment cost.

A 12' x 40' glider or orbiter would provide more operational experience with larger payloads than would a 10' x 20' and hence would make it easier to grow to a larger system should that later be desired.⁴⁶

The elements of a consensus now appeared to be at hand. Following Low's presentation to the Flax Committee, its members felt that NASA had "finally come around to something reasonable"—with the reasonable thing being the glider, not the TAOS booster and orbiter. Such a glider also appeared on the option lists of both Low and the OMB staff. Within the White House, Flanagan met with two of the top executives of North American Rockwell. He told them that there definitely would be a Shuttle program, that the government was about ready to make the decision, but that some issues still had to be sorted out.

Weinberger had already rejected Big Gemini; Low now argued against the glider. Lacking its own propulsion, it could not make use of designs such as TAOS to keep the cost per flight within bounds. Instead, a glider would

46. Memo, Taft and Sullivan to Rice, November 29, 1971.

require its own two-stage rocket, which would drive the cost per flight as high as \$35 million. Low met with Rice, Whitehead, Rose, and Edward David, and wrote that "we demonstrated quite conclusively that a glider would not be cost-effective. Apparently, we also made considerable headway with all the glider advocates in convincing them of this point."

A moment of decision came on December 2, as the OMB sent Nixon a Memorandum for the President. It dealt with space policy, covering a range of issues. The memo included a two-page discussion of the Space Shuttle, and presented the OMB's recommendation: "a smaller reduced cost version of a manned reusable Shuttle with an investment of \$4-5 billion over the next eight years—less than one-half NASA's original proposal." With a stroke of his pen, Nixon would grant his consent:

	<i>Approve</i>	<i>Disapprove</i>
1. Initiate reduced-cost smaller Space Shuttle program	_____	_____
2. Conduct Soviet docking mission	_____	_____
3. Conduct other manned Earth-orbital missions	_____	_____
4. Apollo 16 and 17		
Cancel both missions	_____	_____
Cancel just Apollo 16	_____	_____
Reschedule Apollo 16 and fly both	_____	_____

For NASA, however, this Shuttle that would fit the budget amounted to a throwback to earlier years, when NASA had considered only its own needs and had not yet introduced the large payload capacity that the Air Force would demand. The new shuttle would not be a glider; it would have its own propulsion. It would, however, carry only 30,000 pounds, in a 10 x 30-foot bay. It offered considerably less than the lowest-cost version of TAOS considered to date.

For months, NASA's senior administrators had been negotiating with the OMB, which time and again had pressed for less costly designs. These same administrators then had to turn around and act as an in-house version of OMB, pressing their engineering managers to pursue such designs. These managers operated at the working level; meetings with Weinberger or Flanigan took place far over their heads. They liked the two-stage fully

reusable Space Shuttle; they had nurtured it and did not want to give it up. Low and Myers thus had to put a good deal of effort into keeping their troops in line.

This internal NASA debate, paralleling that between NASA and the OMB, brought such contretemps as Max Faget's reluctance to accept Air Force requirements, the resistance that Heiss and Lindley encountered when they advocated TAOS, and insistence at NASA Marshall that the booster had to be liquid-fueled. The December 2 Space Shuttle, which the OMB proposed and Nixon endorsed a week later, fell outside the bounds of what NASA was willing to take. Fletcher, dealing with both the White House and the OMB, would promptly declare that it was unacceptable.⁴⁷

47. OMB, Memorandum for the President, December 2, 1971; OMB, "Space Shuttle Program," December 10, 1971; Low, Personal Notes No. 60, December 12, 1971; letter, Low to John Logsdon, January 23, 1979.



CHAPTER NINE

Nixon's Decision

Richard Nixon liked space flight. "I can remember Nixon coming off a phone conversation with the astronauts," John Ehrlichman recalls.

And you know, they are up on the moon, and [Nixon was] as high as a kite. He got a big charge out of them. Then when the astronauts would come to the White House for dinner afterwards, he would always be enormously stimulated by contact with these folks. He liked heroes. He thought it was good for this country to have heroes.

Like other presidents before and since, he basked in the reflected glory of spacefarers. When the crew of Apollo 11 returned from the first landing on the Moon, he was aboard the aircraft carrier *USS Hornet* to greet them. He then used this triumph to gain diplomatic advantage, for after hailing the achievement, he set out on a nine-day world tour that took him to capitals in Southeast Asia, India, Pakistan, and Europe. Significantly, he had planned this tour well in advance of the Apollo 11 flight, anticipating its safe return. "The President had rather daringly pegged his trip to the success of this operation," Tom Paine later remarked.

Had he gone out to the Pacific to be present at the splashdown and had there been some kind of an accident, it might have harmed considerably his ability then to have the successful trip, which was his first trip abroad as President. I was scared to death that we would have a fiasco or even a

tragedy. We just wondered whether he knew the odds as well as we did. Well, fortunately Apollo 11 was a success, and in the ensuing world trip, everywhere the President went, the only thing about the United States that anybody wanted to talk about was of course the lunar landing."¹

Yet while Nixon willingly embraced Apollo, which he had inherited from Lyndon Johnson, he took his time in committing the nation to new initiatives, whether in space or in other areas of technology. Between 1960 and 1980, such civilian initiatives were largely a province of Democrats: Kennedy and Johnson with Apollo and NASA, Jimmy Carter with his ambitious synthetic-fuels program in the late 1970s. But when George Shultz presented Nixon with NASA's plan for the space shuttle and urged him to accept it, he did.

Nixon and Technology

If Nixon had wished to emulate Kennedy by supporting a new push in space, he could have endorsed the September 1969 report of the Space Task Group, with its recommended focus on a piloted mission to Mars. Nixon did no such thing. He did not even respond to this report in a timely fashion, waiting nearly six months before issuing his own statement on space policy. Nor did this statement come from a senior advisor such as Ehrlichman or Henry Kissinger. Instead, it was the work of two middle-ranking staffers, William Kriegsman and Clay Whitehead, who reported to Peter Flanigan.

John Kennedy, while in the White House, had repeatedly spoken of space flight with the ring of a clarion call, and it is appropriate to note the contrast. Here is JFK, speaking at Rice University in September 1962:

The exploration of space will go ahead whether we join in it or not, and it is one of the great adventures of all time, and no nation which expects to be the leader of other nations can expect to stay behind in this race for space.

For the eyes of the world now look into space—to the moon and to the planets beyond—and we have vowed that we shall not see it governed by a hostile flag of conquest, but by a banner of freedom and peace.

1. Manchester, *Glory*, pp. 1159-1160; John Logsdon interview, John Ehrlichman, Santa Fe, May 6, 1983, pp. 21, 25-26; Eugene Emme interviews, Thomas Paine, Washington, August 3, 1970, pp. 27-28; September, 1970, pp. 13-14.

We set sail on this new sea because there is new knowledge to be gained and new rights to be won, and they must be won and used for the progress of all people.

But why, some say, the moon? Why choose this as our goal? And they may well ask, why climb the highest mountain? Why, thirty-five years ago, fly the Atlantic? Why does Rice play Texas?

We choose to go to the moon! We choose to go to the moon in this decade, and do the other things, not because they are easy, but because they are hard. Because that goal will serve to organize and measure the best of our energies and skills. Because that challenge is one that we are willing to accept, one we are unwilling to postpone, and one we intend to win.²

Similarly, here is Nixon in his statement of March 1970, which amounted to a most uncertain trumpet:

Having completed that long stride into the future which has been our objective for the past decade, we now must define new goals which make sense for the Seventies. We must build on the successes of the past, always reaching out for new achievements. But we must also recognize that many critical problems here on this planet make high priority demands on our attention and our resources. By no means should we allow our space program to stagnate. But—with the entire future and the entire universe before us—we should not try to do everything at once. Our approach to space must continue to be bold—but it must also be balanced....

We must realize that space activities will be a part of our lives for the rest of time. We must think of them as part of a continuing process—one which will go on day in and day out, year in and year out—and not as a series of separate leaps, each requiring a massive concentration of energy and will and accomplished on a crash timetable.... We must also realize that space expenditures must take their proper place within a rigorous system of national priorities.

The statement endorsed the activities that were under way at the moment or were well along in preparation: additional Apollo flights, Skylab, planetary

2. *New York Times*, September 13, 1962, p. 16.

missions, and Earth-orbiting spacecraft. It called for further moves toward international cooperation in space. It stopped, however, well short of endorsing the Space Shuttle:

We should work to reduce substantially the cost of space operations [emphasis in original]. Our present rocket technology will provide a reliable launch capability for some time. But as we build for the longer-range future, we must devise less costly and less complicated ways of transporting payloads into space. Such a capability—designed so that it will be suitable for a wide range of scientific, defense and commercial uses—can help us realize important economies in all aspects of our space program. We are currently examining in greater detail the feasibility of re-usable space shuttles as one way of achieving this objective.

This paragraph amounted to a Nixonian blessing for the continued "examining" that would proceed during the next two years. The overall statement, however, endorsed only one new initiative: "Grand Tour," a program that would take advantage of a rare alignment of the outer planets to visit Jupiter, Saturn, Uranus, and Neptune. Moreover, Nixon's statement specifically supported his budget for FY 1971, which continued a policy of cuts in appropriations that dated to 1966. In 1970, NASA was still in retreat, and this statement underscored this march to the rear.³

Yet amid this retreat, Nixon maintained strong support for his existing program, with the SST being a prime example. In Ehrlichman's words,

Nixon died very hard on the SST, and he had a commitment to that which had to do with chauvinism. I think, is the proper word. We had to be at the leading edge of this kind of applied technological development. And if we weren't, then a great deal of national virtue was lost, and our standing in the world and all that. He was terribly troubled to go to an international conference and have the French president arrive in an SST. That was why that was very hard on him.

In an attempt to recoup, Nixon borrowed an idea of the Democrats: that the government should use the resources of aerospace to solve domestic

problems. During the second half of 1971, several of his senior advisors tried to launch an effort called the New Technology Opportunities Program (NTOP), which sought to define specific projects that might be ripe for federal support. The key man in this effort, William Magruder, had been Nixon's head of the SST program. Ehrlichman recalls that Nixon gave an instruction: "Let's keep in science and technology, and let's find something good for Magruder to do."⁴

The activity began on July 1, when Ehrlichman sent letters to 15 agencies, asking for proposals. The responses went to the Office of Science and Technology, where Edward David's staff carried out initial evaluations. Then in September, Nixon bypassed David as head of this program, even though David had come up within Bell Labs as a technologist, and even though his purview specifically included technology. In an action that effectively downgraded this side of David's domain, Nixon named Magruder as head of NTOP. The OST staff continued its assessments, but Magruder broadened the review to include people from the Treasury Department and the Council of Economic Advisers. Magruder also worked with the OMB's Donald Rice, who assigned a staff member, Hugh Loweth, to work full-time on budgetary aspects.

The effort went forward under tight deadlines, for Ehrlichman wanted to have a finished set of proposals in hand by the end of 1971, in time for inclusion in the FY 1973 budget and the President's State of the Union message. Magruder expanded his reach by seeking ideas from private industry, sending out hundreds of letters to corporations and trade associations. "We were continually in a crisis situation," said one OST manager. "Toward the end, we were killing those guys in the OMB, hitting them with more and more proposals every day. Poor Hugh Loweth was working practically a 24-hour day."

The reviewers quickly discovered, however, that the prospective domestic initiatives carried difficulties that ranged well beyond the merely technical. One important proposal called for full-scale development of high-speed rail transportation in the Northeast, laying new rail and refurbishing passenger stations. The New York Central and Pennsylvania railroads, recently merged into the Penn Central, had allowed this service to deteriorate badly; the Penn Central had gone bankrupt in 1970. Yet its tracks and rights-of-way were still serviceable.⁵

3. Nixon, "Statement by the President," March 7, 1970. Reprinted in Launius, *NASA*, pp. 216-221.

4. John Logsdon interview, John Ehrlichman, Santa Fe, May 6, 1983, pp. 2-4.

5. *National Journal*: October 23, 1971, pp. 2114-2124; October 30, 1971, pp. 2156-2163; May 6, 1972, pp. 756-765; *Audacity*, Summer 1993, pp. 52-62.

We pushed that pretty hard," said Lawrence Goldmuntz, who directed the OST's proposal evaluations.

And who can argue that it shouldn't be a high-priority item? But in analyzing that proposal the White House also had to take into account the fact that there are several thousand government jurisdictions involved, that the Penn Central is not the most popular railroad in the country today, that it might get athwart union work rules—and well, a number of complicated issues like this came up.

Another proposal envisioned the development of two-way television, which would foreshadow today's personal computers with e-mail. Two-way TV would allow individual citizens to communicate directly with city social-service agencies, including health, welfare, and police-protection programs. "But we quickly got caught in a crossfire between the Corporation for Public Broadcasting, the Office of Telecommunications Policy, and the cable TV interests," said Goldmuntz. "The policy questions were just too complex."

Another concept promised to develop integrated utilities, which would combine sewage and solid-waste disposal, power, heat, and light within a single system. Such systems, built as modules, offered major savings through lower fuel consumption, with the modules being installed in office towers and apartment buildings. Such integrated services, however, would have raised opposition from unionized municipal workers.

Other proposals ran afoul of recent changes in the national mood. The Atomic Energy Commission had a long-running interest in peaceful uses of nuclear explosives. Its officials endorsed a demonstration project that would use multiple detonations to fracture impermeable rock formations that held natural gas. The concept gained high marks from Magruder's reviewers; the AEC plan even seemed to promise commercial feasibility. However, 1972 was an election year, and with environmentalists showing their strength, the Administration could not touch it.

Still other proposals faced political opposition, such as a plan to build offshore terminals for deep-draft supertankers that drew too much water to enter conventional ports. Such terminals promised to cut shipping costs by eliminating the need to route the supertankers to the Caribbean, where they would transfer their cargoes of oil to smaller tankers of lesser draft. This proposal,

however, faced strong opposition from governors of states in the Northeast, who feared oil spills. It also would have tended to push the White House toward endorsing expansion of oil imports, a policy that Nixon was reluctant to support.

Four senior White House officials carried out the final review: John Ehrlichman, head of the Domestic Council; OMB director George Shultz; Peter Flanigan, special assistant to the President; and Peter Peterson, director of the Council on International Economic Policy. They declined to endorse any of Magruder's proposals, and the main reason was that in the course of the NTOP exercise, key people had come to realize that they truly knew little about the process of technological innovation. By Christmas, NTOP was moribund.

"We did think in the summer that we could do more and do it quickly," said Peterson. "By December, we were determined to go slow and keep our feet on the ground. I didn't think we should jump into anything before we knew where we were going." Edward David added that "one of the things many of us had driven home more clearly than before was that R and D is not the whole story—you've got to take into account customs, mores, politics, existing structures, and a whole host of other things when you attack a technological issue." Raymond Bisplinghoff, deputy director of the National Science Foundation, concluded that this exercise "verified that we do not know how to make major interventions by the federal government in the R and D sector."

It is a matter of record that as the NTOP fell, the Space Shuttle rose and won Nixon's approval. The latter event, however, did not follow from the former. Edwin Harper, assistant director of the Domestic Council, told *National Journal*, "I was at all the relevant meetings and the two programs were never discussed in terms of a trade-off. The timing of the space-shuttle decision had an independent history." Both NTOP and the shuttle had to stand on their respective merits; in the end, only the Shuttle survived the cut.

NTOP nevertheless was important, for it represented a serious White House attempt to redirect the resources of aerospace toward new domestic priorities. When the attempt faltered, it soon became clear that Nixon would not try to help the beleaguered aerospace industry by having its people work on mass transit or pollution control. Instead, he would give them an election-year gift by keeping that industry's resources within the realm of aerospace.⁶

6. *National Journal*, May 6, 1972, pp. 756-765.

Space Shuttle: The Last Moves

In mid-November, three weeks after the Director's Review and three weeks before the OMB would request Nixon's approval for a downsized shuttle, Low wrote: "The shuttle configuration is beginning to be focused on a considerably smaller orbiter with external hydrogen and oxygen tanks (but with the same payload size and weight) and with a pressure-fed recoverable booster which might be parallel staged." The use of a pressure-fed booster was an option dating to the studies of early summer; the possibility of parallel staging, with booster and orbiter engines all burning at liftoff, showed the influence of TAOS. On November 22, in his report to Rice, Low was more definite: "The most promising candidate configuration today is the Mark I/Mark II orbiter with the parallel-staged pressure-fed booster."

Ten days later, the OMB's Memorandum for the President acknowledged NASA's recent design revisions but called on the agency to accept a version of the shuttle that would be less costly still:

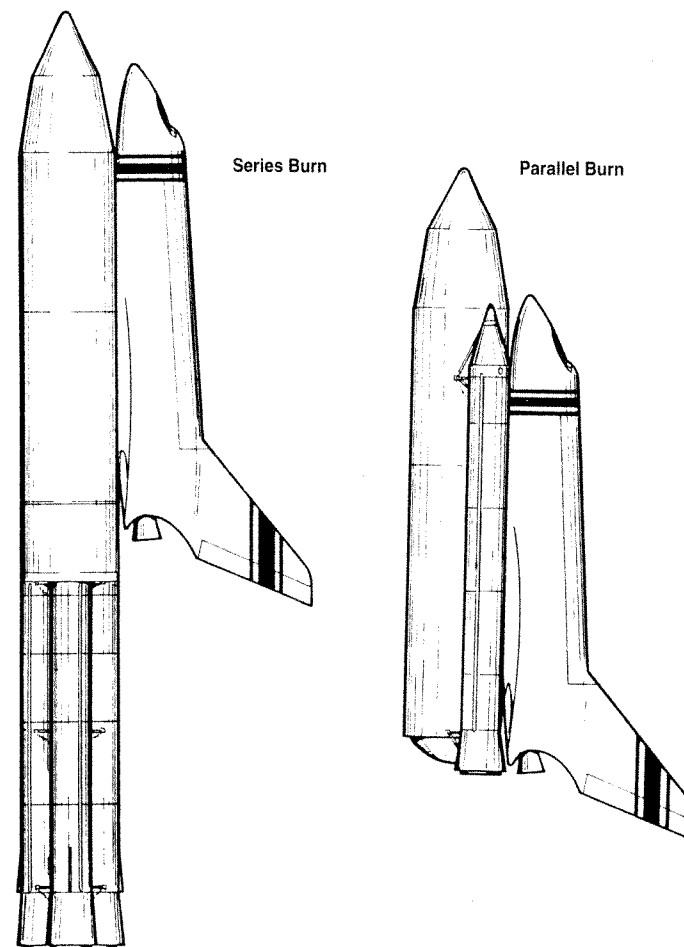
Last year NASA was proposing a \$10-12 B Shuttle. In response to questions from OMB and OST about whether the benefits justified such a large investment, NASA has since designed a \$6 B Shuttle which can do all the missions of the larger, more expensive one because it has exactly the same payload capability. (We think both costs are underestimated, perhaps by 50%, i.e., cost overruns are likely on both but more likely on the more expensive version.)

In either case, NASA would plan to replace all of the U.S. expendable booster programs with the Shuttle. Thus, one program, the Shuttle, would dominate NASA for the coming decade, as did Apollo in the 1960's. This would make efforts to reorient NASA to domestic pursuits more difficult, and tend to starve unmanned earth applications missions for resources.

The Shuttle alternative that is chosen must balance costs, benefits and subjective considerations.

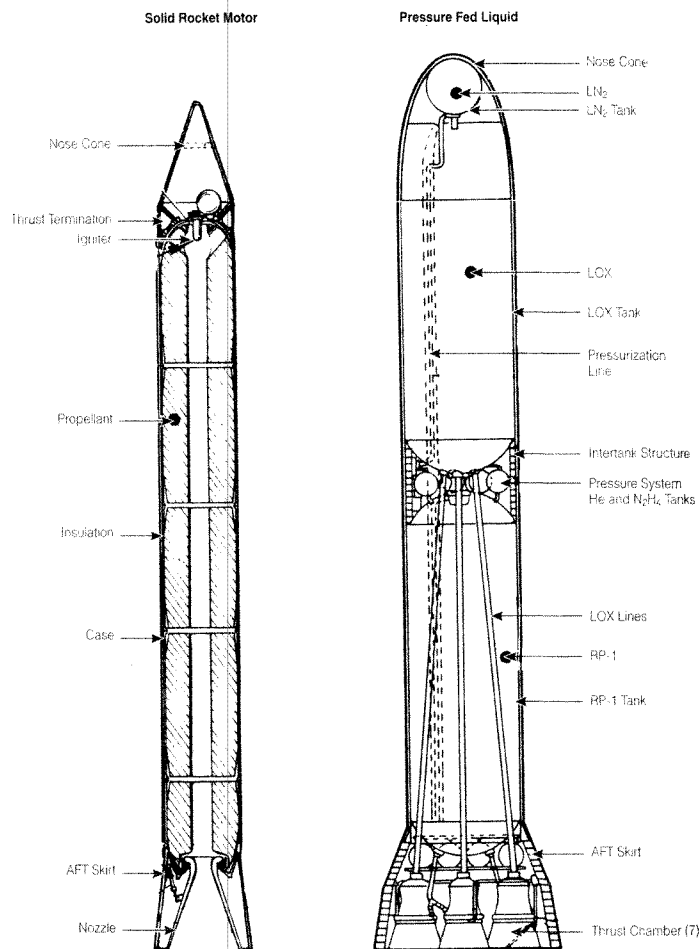
What are the Options? NASA, NASA contractors, OST, PSAC and OMB have all given consideration to alternatives to NASA's large Space Shuttle proposal. In summary these alternatives run the gamut from:

- large systems with both reusable powered orbiters and boosters (\$12 B) to*
- small systems with an unpowered reusable orbiter and a non-reusable launch vehicle (\$3 B).*

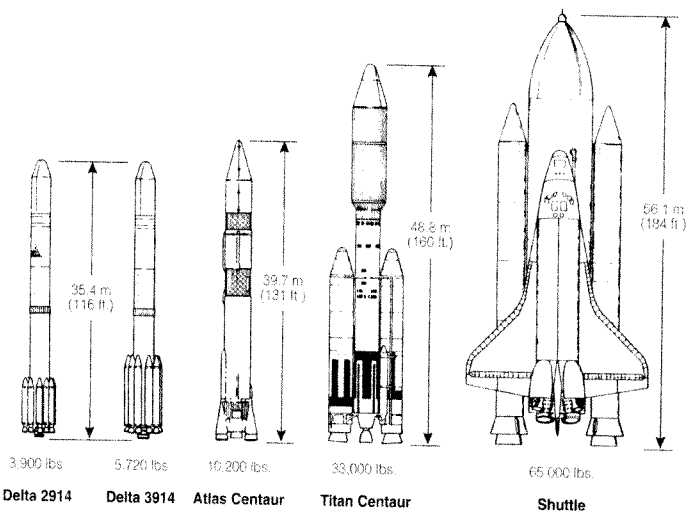


Series vs. parallel burn. Series burn calls for conventional staging, with the orbiter engines igniting following cutoff of the first stage. Parallel burn ignites both booster and orbiter engines prior to liftoff. TAOS concepts called for parallel burn. (Thiokol)

Space Shuttle Booster Candidates



TAOS alternatives: solid rocket motors and pressure-fed liquid boosters, to fly as strap-ons.
(Thiokol)



Expendable launch vehicles of the 1970s, intended for replacement by the Shuttle. (NASA)

Operating costs vary from a high of \$30 million per launch for the lowest investment cost option to a low of \$5 million per launch for the highest investment cost option.

The revised program proposed in this memorandum would develop a smaller reduced cost version of a manned reusable Shuttle with an investment of \$4.5 billion over the next 8 years—less than one-half NASA's original proposal.

The OMB proposal was not the result of a thoughtless exercise aimed at pressing NASA until the pips squeaked; it represented a serious alternative. With its 10 x 30-foot payload bay and 30,000-pound capacity, the OMB shuttle would still capture some 80 percent of the payloads of the larger designs. Similarly, the OMB would not permit its shuttle to be all things to all people; its memo to Nixon stated that "for national security purposes, we may not want all our eggs in one basket." The OMB stated explicitly that the

nation was to "retain the reliable Titan III expendable booster to launch the few largest payloads that would not fit the smaller Shuttle. These include space telescopes and large intelligence satellites."⁷

Nixon took about a week before he read and accepted the OMB memo. On Saturday, December 11, Fletcher and Low met with Rice, David, and Flanigan. These NASA officials learned that Nixon indeed had decided to go ahead with the shuttle—provided that it was the downsized version of the OMB. This brought considerable and heated discussion. Fletcher finally declared that he could not accept such a decision, and that he wanted to see the President. This meeting did not take place, for Fletcher subsequently decided that he would not help his cause by staging a confrontation with Nixon.

Fletcher nevertheless would fight for a larger shuttle, and in doing this, he would stand for NASA in the fashion of his predecessors. Paine had been the visionary, pushing toward Mars. Low had succeeded Paine; as Acting Administrator, he pushed strongly for two-stage fully-reusable designs. He also sought OMB approval to initiate shuttle development during FY 1972, and won permission to proceed with the SSME on such a schedule. By contrast, Fletcher had acquiesced to OMB pressure from the outset, abandoning the high-cost alternatives as he struggled to meet the OMB's stringent cost limits. He nevertheless had a limited amount of wiggle room, and he would make the most of it.

Though the OMB memo to Nixon specified a development cost, it did not state a payload size or capacity. The OMB presented its numbers separately in a paper for NASA: 10 x 30 feet, 30,000 pounds. Low responded quickly; on Monday, December 13, he sent a memo to Dale Myers that asked for an in-house assessment of this OMB shuttle, to be completed by the end of the month. In this assessment, Myers was to try once more for the full-capacity Shuttle of 15 x 60 feet and 65,000 pounds, by comparing its merits with those of the OMB's configuration.

Flanigan also proved helpful, as he sent a memo to Fletcher from the White House: "None of the figures in the paper given to you are set in concrete. Rather, they should be viewed as a new way to approach the problem, against which an initial estimate will be made within a couple of weeks. Obviously,

those figures will have to be refined in succeeding weeks." He added pointedly, "There is no written directive from the President on this subject."⁸

Fletcher, however, had to prepare once again to give ground. The OMB was willing to permit NASA to build its shuttle with the SSME: it was willing to bypass the interim Phase I orbiter with its J-2S engines. The OMB also would allow a booster "of the reusable pressure-fed type," which Low had recommended. Nevertheless, while rejecting the OMB's small payload size and weight, Fletcher and Low had to find new ways to save money and to cut the developmental cost anew.

With OMB ready to toss the Air Force's payload bay out the window, and with the OMB calling for continued use of the Titan III for large Air Force payloads, it was appropriate to take a fresh look at another Air Force requirement, which had demanded a crossrange of 1,100 nautical miles. This had been a prime reason for the choice of a delta wing for the orbiter, which drove up the weight of the Shuttle and increased its cost by requiring more thermal protection. With the OMB now pressing NASA to return to the payload capacity recommended by Max Faget, the agency had to consider whether it might cut costs by also returning to Faget's straight-wing orbiter design.

Charles Donlan, acting director of the Shuttle Program Office, ruled this out. In a memo to Low, he emphasized that high crossrange would be "fundamental to the operation of the orbiter." It would make the Shuttle maneuverable, greatly broadening the opportunities to abort a mission and perhaps save the lives of astronauts. A high crossrange would also afford frequent opportunities to return to Cape Canaveral in the course of a normal mission, following launch from that site.

Delta wings also promised advantages that were entirely separate from crossrange. A delta orbiter would be stable in flight from hypersonic to subsonic speeds, throughout a wide range of nose-high attitudes. The aerodynamic flow over such an orbiter would be smooth and predictable, with the delta wing thus permitting accurate predictions of heating during reentry and giving confidence in the design of the thermal protection system. In addition to this, the delta vehicle would experience relatively low temperatures of 600 to 800 °F over its sides and upper surfaces.

7. Low, Personal Notes No. 58, November 14, 1971; letter, Low to Rice, November 22, 1971; OMB, Memorandum for the President, December 2, 1971.

8. OMB, "Space Shuttle Program," December 10, 1971; Low, Personal Notes No. 60, December 12, 1971; memos: Low to Myers, December 13, 1971; Flanigan to Fletcher, December 17, 1971; *Science*, May 30, 1986, p. 1103.

By contrast, straight-wing configurations would produce complex hypersonic flow fields, with high local temperatures and severe changes in temperature on the wing, body and tail. Temperatures on the sides of the fuselage would run from 900 to 1,300 degrees, making the analysis and design of thermal protection more complex. During transition from supersonic to subsonic speeds, the straight-wing orbiter would experience unsteady flow and buffeting, making it harder to fly.

Because of this combination of aerodynamic and operational advantages, Donlan favored the delta wing for reasons that were entirely separate from those of the Air Force. Fletcher, however, still could back off on the issue of payload-bay capacity. A smaller bay would give a smaller and somewhat less costly orbiter. Reducing the payload weight would trim the size and mass of the entire Shuttle, cutting costs even further. Moreover, for the past several weeks Fletcher had been holding in reserve the possibility of such cuts.

He had gone to lunch on October 19, 1971, with David Packard, the Deputy Secretary of Defense, and Packard had stated that he felt very uneasy over the Air Force's design requirements for the shuttle. As Fletcher wrote in a memo to Low,

The requirements which he was particularly concerned about were the cross-range requirement, and payload requirement, and the size requirement. He felt that the cross-range requirement might have been an artificial one and although he didn't fully understand the implications of it felt that if it were causing difficulties, it could easily be modified. I assured him that the diameter requirement came primarily from NASA and not from the Air Force, but that the length probably came from the Air Force. He knew quite well which program caused the length difficulty and although it can't be discussed in this memorandum, it is clear to both of us that something could be done in this regard also. We both agreed that the payload requirement was somewhat arbitrary at this point.

Fletcher did not discuss this with Rice or Weinberger, for he was about to engage them in the high-stakes game that followed the Director's Review, and he did not want to tip his hand. Two months later, however, with Fletcher preparing to play this card of shuttle capacity, Low's memo of December 13 directed Myers to compare four cases: the OMB shuttle, the full-size NASA

shuttle, a design of 12 x 40 feet and 30,000 pounds that NASA had studied previously, and a new case of 14 x 50 feet and 65,000 pounds. They all knew, however, that they could not start with a smaller bay and then enlarge it. Donlan had written that such a course "is not considered practical. The most cost effective system is one sized properly at the outset for its intended use."⁹

In addition to this, with Low presenting new alternatives to the NASA staff for evaluation, Don Rice found himself in a position to add a few alternatives of his own. He had formal channels for receiving information from NASA, but now he opened up some back-channels that gave him access to additional sources. "Some of my information came from the Defense Department, but not very much of it," he later remarked. "Some of it came from industry. There were clearly some people in industry who were concerned that NASA was going to lead them down the road of another C-5A debacle and that they would end up with nothing."

Fletcher saw what was happening:

He would come up with his own drawings on what it should look like. He had private sources that he was turning to. Contractors wanting their particular version. Contractors do that. And there is no reason why they shouldn't. I hate to say this about Don but he really didn't understand aerospace contractors. Tell him to design a cheaper system, he'd design a cheaper system. He went that far; he came up with his own design.

Fletcher took it in stride: "We kept dealing with him. After all, you have to be polite, because you never know who's going to be your boss next." Low also joined in the exchanges. At one point, Rice presented a developmental cost of \$4 billion and claimed that it came from a contractor whose name he could not divulge. Low asked him to disclose the contractor, so that NASA could critique his claim. Rice refused, but based on the questions he had asked, Low concluded that the contractor was North American Rockwell. That company had given far more information to NASA than to Rice, and Low determined that Rice's \$4 billion left out company profit and the cost of NASA program support. Indeed, it was the equivalent of numbers that NASA itself had shown.

9. Memos: Fletcher to Low, October 20, 1971; Donlan to Low, December 5, 1971; Low to Myers, December 13, 1971.

Every time Rice got a bright idea, he would send it over to NASA for assessment by its staff. The OMB's William Niskanen, head of its Evaluation Division, chimed in with similar requests of his own. Within NASA, William Lilly was the chief budget officer and was close to the working level where people had to take the time to deal with these matters.

Lilly tried to come between Rice and his staff:

It was some dirty warfare going on at that time. Our approach was to, in essence, to create doubt in their boss's mind of the value of the work they had done. We did our studies and in many ways in order just to prove that they were biased, and they were wrong and they didn't know what they were doing. And any way we could embarrass them, we did it.

Lilly took care to control the flow of information from NASA to the OMB, for he knew that a small leak can sink a great ship:

I reached the point, to hell with you, back of my hand to you. I mean, you can't do your job if I don't let you in the agency [NASA]. And in essence, I probably went about shutting them off from information throughout the agency. I just told my people that, in a way that they understood, but not really telling them, they weren't to respond. I wanted to see everything that went over. There was to be no answers to questions given throughout the agency to OMB without it coming through me personally.

"They kept throwing road blocks on us," Lilly continues. "Every time we would come up with a study and do what they wanted in terms of comparative analysis. They couldn't challenge what we had done in many ways, they would turn around and come with studying a new concept." In time, Niskanen sent a request that went too far: "To be brutally frank about it, the reaction we had or my attitude on it, and what, in essence, I told them? I told them to go shove it up your ass. I wasn't going to do any such thing. It didn't make any goddamn sense. I wasn't about to do such a study, and we never did."

Fletcher, by then, was using channels of his own: "We were kind of mean to Don in going over his head all the time. But I figured that was the only way we could get this thing done." He met with Caspar Weinberger "quite fre-

quently," with Peter Flanigan "almost as often," and with George Shultz "two or three times." He also arranged a meeting with John Ehrlichman.¹⁰

These exchanges continued through Christmas. On Monday, December 27, having reviewed the latest contractor studies, Fletcher prepared to play his poker card of shuttle capacity and to make an offer that would counter the OMB shuttle of two weeks earlier. "We met all day to discuss the various options," Low wrote. Low continued those meetings on Tuesday morning, talking with Myers, with Lilly, and with others as well. Following those meetings, Low and Fletcher agreed on the terms of their counteroffer. Fletcher then presented these terms to Weinberger in a December 29 letter:

We have concluded that the full capability 15 x 60' - 65,000# shuttle still represents a "best buy", and in ordinary times should be developed. However, in recognition of the extremely severe near-term budgetary problems, we are recommending a somewhat smaller vehicle—one with a 14 x 45' - 45,000# payload capability, at a somewhat reduced overall cost.

This is the smallest vehicle that we can still consider to be useful for manned flight as well as a variety of unmanned payloads. However, it will not accommodate many DOD payloads and some planetary payloads. Also, it will not accommodate a space tug together with a payload, and will therefore not provide an effective capability to return payloads or propulsive stages from high "synchronous" orbits, where most applications payloads are placed.

An accompanying table compared costs for five options:

CASE	1	2	2A	3	4
Payload bay (ft.)	10 x 30	12 x 40	14 x 45	14 x 50	15 x 60
Payload weight (lbs.)	30,000	30,000	45,000	65,000	65,000
Development cost (\$billions)	4.7	4.9	5.0	5.2	5.5
Operating cost (\$millions/flight)	6.6	7.0	7.5	7.6	7.7
Payload costs (\$/pound)	220	223	167	115	118

10. John Logsdon interviews: Donald Rice, November 13, 1975, p. 2; James Fletcher, September 21, 1977, pp. 20-21, 23; John Mauer interview. William Lilly, Washington, October 20, 1983, pp. 28-33; Roger Launius interview. James Fletcher, September 19, 1991, p. 14; Low, Personal Notes No. 61, January 2, 1972.

This table included the four cases cited by Low in his memo to Myers of December 13, and reviewed by Myers. Case 1 was the OMB shuttle; Case 4 was the shuttle NASA wanted and now apparently would not have. Following standard custom, the option Fletcher proposed was right in the middle as Case 2A.¹¹

Fletcher privately knew that he could go still lower. Talking with Low, he decided that they could accept something as small as 14 x 40 feet with 40,000 pounds. The two men then went to an afternoon meeting with White House and OMB officials: Shultz, Weinberger, Flanigan, David, Rice and Rose. Shultz was now the key man; he headed the OMB, he was Rice's boss, and he had Nixon's ear.

Shultz looked at NASA's presentation and decided that the only thing that made any sense, as NASA had said all along, would be the full-size version, Case 4. Shultz, however, did not press this point for Rice objected vigorously. Rice's staff was still active; only one day earlier, his economist John Sullivan had sent him a memo arguing anew that the most cost-effective system was still the Titan III. The meeting broke up with no decision. Fletcher and Low, however, came away fairly confident that they would at least get Case 2A, which they had recommended. Indeed, Shultz's support, however tentative, allowed them to hope that they might even win the full-capability Case 4.

Rice again prevailed, as he talked further with Weinberger. In a phone conversation with Fletcher, Weinberger stated that he wanted NASA to look at a 14 x 45-foot shuttle—with 30,000 pounds of payload, only two-thirds that of Case 2A. In Low's words, "Fletcher came close to telling Weinberger to go to hell but restrained himself perhaps better than I could." Fletcher then phoned Shultz and talked with him at length. Shultz remained unwilling to make a decision, but recommended that NASA should take one more look at Rice's request.

Although Rice was holding firm on a weight of 30,000 pounds, he now was willing to budge slightly on payload size, for Sullivan's memo had discussed a 12 x 40-foot shuttle with twin solid boosters. Though this configuration would carry no more weight than a Titan III, it could fly with the boosters of a Titan III: 120-inch solid motors that were in production and had known costs. Such a shuttle still would not match the cost-effectiveness of the Titan III itself, but it would come close. In Sullivan's own analysis, that

Titan would save only \$100 million when compared to that shuttle. Within the OMB, this was as near to an endorsement as any type of shuttle was likely to receive.

Low phoned Rice and asked him to put his questions in writing. Rice replied that he might have further questions subsequently, but he drew up a set of queries and sent it over to Low late on Friday evening, which was New Year's Eve. Low discussed them with Fletcher and Myers over the weekend; on Monday, January 3, they completed their response. A sampling will illustrate the exchanges:

1. If future budgets for NASA were constrained to \$3.2 - \$3.3 B, would you want to do a large shuttle?

Answer: The answer is yes. The NASA budget is not committed to exceed the FY 1973 level even if the maximum credible cost overrun occurred.

2. Why should a relatively few space station modules for the mid-1980's determine the size and weight capabilities of the shuttle? What other missions are really driving the payload size and weight requirements?

Answer: In addition to "a few space station modules," the payload length is driven above 40 feet by most NASA planetary payloads, most DOD payloads, and a few of the NASA science payloads.... The payload diameter is driven above 13 feet by manned payloads, some NASA science payloads, and some NASA planetary payloads. The payload weight is driven above 40,000 lbs. not only by space station modules but by space station resupply logistic vehicles, as well as sortie cans;¹² a 40,000-lb. payload capability is also exceeded by many DOD payloads, as well as 13 different science, applications and planetary payloads.

4. What capabilities and dollar benefits would be lost by going to a 12 x 40' (30 - 35,000 lb.) shuttle launched by SRM's?

Answer: At this size and weight we lose most DOD payloads, all manned payloads (including resupply logistics vehicles), most planetary payloads, and many science and applications payloads.

11. Letter, Fletcher to Weinberger, December 29, 1971; Low, Personal Notes No. 61, January 2, 1972.

12. Small laboratories that would not fly freely in space but would remain within the payload bay and return with the Shuttle orbiter.

These exchanges with Rice would feed into a letter that Fletcher intended to prepare for Weinberger on Monday, January 3. On Sunday, however, with no resolution in sight, Low confided privately that "there is nobody in the White House willing to make any decisions."

He later wrote,

*The single most significant factor was that there was no top-level leadership in the White House. Nixon was unwilling to deal with his agency heads and dealt solely with his staff. This placed a great deal of the decision-making responsibility with the OMB, and by definition the OMB is far more interested in short-range budgetary problems than in the long-range future of the nation.*¹³

The Hinge of Decision

Low did not know it, but the commitment he wanted already was imminent. At a recent budget session held in Key Biscayne near Miami, a decision had been made to include funds for the Shuttle. A subsequent decision called for Nixon to release an announcement at the Western White House in San Clemente, California. He would also meet there with Fletcher, who would hold a press conference. Fletcher learned of this during that New Year weekend. He and Jonathan Rose responded by asking William Anders of the National Aeronautics and Space Council to prepare a draft of this presidential statement.

Monday dawned, with the day's action items prominently featuring Fletcher's letter to Weinberger and NASA's responses to Rice's question set. People at the Manned Spacecraft Center had been working over the weekend to come up with the answers, but when these responses reached Washington, early that afternoon, they were quite inconsistent. It quickly developed that some of the specialists in Houston had misread some key tables of data. With time pressing, NASA Headquarters gave them only one hour to come up with the right answers, which they did. These answers made it possible for Fletcher to finish his letter:

Our view that the shuttle with the 14x45 - 45,000# capability is the minimum acceptable option is still valid.

The OMB proposed option of a 14x45 - 30,000# shuttle is not acceptable because it will not handle manned space station modules, manned sortie flights or manned resupply missions in a standard space station orbit; in short, it would not provide a manned spaceflight capability for the United States. Also, this shuttle would not handle 28 different science, applications and planetary payloads that could be carried with a 45,000# payload capability.

At 6 p.m., Fletcher and Low met in George Shultz's office with a group of participants that again included Shultz, Weinberger, Flanagan, Rice, and David. The opponents' positions remained the same, with Rice calling for a smaller design and David proposing that they put off a commitment for several months, to give NASA time to refine its most recent estimates. Shultz responded differently. Unlike those critics, he had been at Key Biscayne.

He had given Rice full opportunity to raise his objections. Now, in this meeting, he had seen NASA respond to Rice's questions, with answers that militated strongly against the OMB shuttle. What happened next proved to be the hinge of decision.

Only Low wrote of this moment soon afterward, and in a terse manner: "Shultz agreed with our position that the 15 x 60' 65,000 lb. Shuttle should be developed." Weinberger, Rice, and Fletcher all gave interviews during subsequent years, but did not discuss the events of this meeting. Fletcher and Low, however, spoke of the matter with senior NASA officials, who would present their own recollections:¹⁴

Lilly: Rice got a little bit confused. There was a feeling with Low and Fletcher that Rice got too carried away, moving toward misstatements, trying to exaggerate some things. George Shultz picked up the phone and called Morgenstern during the meeting and asked him about it. Now, NASA had prepared for this kind of thing—to be sure Morgenstern was fully knowledgeable. And Shultz got off the phone and made the decision—we'll go this way, and to prepare the papers for the approval.

13. Memo, Sullivan to Rice, December 28, 1971; "Space Shuttle Questions Provided by OMB on 12/31/71"; Low, Personal Notes No. 61, January 2, 1972; letter, Low to John Logsdon, January 23, 1979.

14. Memos: David N. Parker (White House staff), December 31, 1971; Rice to Shultz, January 3, 1972; letter, Fletcher to Weinberger, January 3, 1972; Low, Personal Notes No. 62, January 15, 1972.

Donlan: *I get this directly from Dr. Fletcher. He was telling his wife the story of how he went over there with the information that I supplied him from our shuttle program studies that showed that there wasn't all that much to be gained by going 45-foot length and, furthermore, it invalidated a lot of missions that the Air Force claimed they needed for the sizing of the satellites. So Shultz said, "Well, what are you fooling around with that 45-foot configuration for? It doesn't cost that much more. Why don't you get the one you want—take the 60-foot one." And Fletcher came back with that message. That's how it was settled.*

Willis Shapley, NASA Associate Deputy Administrator: *George Shultz—he had personally spent almost as many hours going through all the planning studies, and especially this famous economic analysis that Klaus Heiss at Mathematica had done. And he personally called up Oskar Morgenstern and other people there, and he was satisfied, finally, that it was a reasonable proposition. So when it was clear that all the boys made their case, Shultz said, 'If we're going to do it, let's do it right; let's do the big shuttle and forget about the Bureau of the Budget shuttle.' So that's how we ended up with what it was."*

It was Christmas in January, for whereas Fletcher and Low had come in hoping for approval of the downsized version that they really did not want, Shultz now was ready to recommend that they receive the full-size version that had not been in play for over a month. Similarly, the views of Rice and David would carry no weight. Shultz, after all, was Rice's boss, and it was Shultz who would meet with Nixon. David also reported to Nixon, but he had little clout. He had been bypassed in favor of Magruder during the recent NTOP exercise. A year later, amid a post-election White House reorganization, Nixon would abolish his post of presidential science advisor—and would also abolish the OST in the bargain.¹⁵

The decision to proceed with the Shuttle became firm during the meeting in Shultz's office, with Shultz confirming his assent to NASA's request for \$200 million as startup funds for the Shuttle within Nixon's FY 1973 budget.

¹⁵ John Mauer interviews: Charles Donlan, Washington, October 19, 1983, pp. 33-34; William Lilly, Washington, October 20, 1983, p. 39; Willis Shapley, October 26, 1984, p. 26; *Science*, January 19, 1973, p. 233; February 2, 1973, pp. 455, 458-459; February 16, 1973, p. 641; March 30, 1973, p. 1311.



Nixon meets with Fletcher and approves the Shuttle as a program, January 5, 1972.
(National Archives)

Flanigan now asked Low and Fletcher to prepare a draft of the presidential statement—which Anders was writing already. In turn, Anders' statement formed the bulk of the material used in Nixon's release.

As recently as November, Flanigan had anticipated that any White House announcement would be low-key. At that time, with the \$3 billion glider as the likely new initiative, Flanigan expected to see the main coverage limited to the aerospace trade press, thereby reassuring this industry of Nixon's support while avoiding the high visibility that would draw fire from critics. The Space Shuttle, however, had metamorphosed now into a \$5.5 billion program. As early as the previous Friday, prior to the meeting in Shultz's office, a White House staffer had already laid out the high-profile announcement that now was scheduled for Wednesday, January 5, 1972.

Fletcher and Low flew out to California, editing two NASA statements along the way. Nixon greeted them at the Western White House, as did John

Ehrlichman. Though the President had planned to spend only 15 minutes with his visitors, the meeting ran well beyond a half-hour as he showed strong interest in the Shuttle and the space program. Fletcher had brought a model of a TAOS, and Ehrlichman would remember "Nixon's fascination with the model. And he held it and, in fact, I wasn't sure that Fletcher was going to be able to get it away from him when the thing was over."

Nixon stated that NASA should stress civilian applications but should not hesitate to note the military uses as well. He showed interest in the possibility of routine operations and quick reaction times, for he saw that these could allow the Shuttle to help in disasters such as earthquakes or floods. He also liked the idea of using the Shuttle to dispose of nuclear waste by launching it into space. Fletcher mentioned that it might become possible to collect solar power in orbit and beam it to Earth in the form of electricity. Nixon replied that such developments tend to happen much more quickly than people expect, and that they should not hesitate to talk about them.

He liked the fact that ordinary people, who would not be highly-trained astronauts, would be able to fly in the Shuttle. He asked if the Shuttle was a good investment, and agreed that it was indeed, for it promised a tenfold reduction in the cost of space flight. He added that even if it was not a good investment, the nation would have to do it anyway, because space flight was here to stay. Fletcher came away from the meeting saying, "The President thinks about space just like McCurdy does," referring to a colleague within NASA's upper management.

Although his formal statement largely reflected NASA's views, Nixon edited the draft in his own hand. The final version showed a firmness and sense of direction that had been utterly lacking in his March 1970 statement on space policy. It also featured a grace note that might have suited John Kennedy:

I have decided today that the United States should proceed at once with the development of an entirely new type of space transportation system designed to help transform the space frontier of the 1970s into familiar territory, easily accessible for human endeavor in the 1980s and '90s.

This system will center on a space vehicle that can shuttle repeatedly from earth to orbit and back. It will revolutionize transportation into near space, by routinizing it. It will take the astronomical costs out of astronautics. In short, it will go a long way toward delivering the rich benefits of practical

Nixon's Decision

space utilization and the valuable spinoffs from space efforts into the daily lives of Americans and all people....

Views of the earth from space have shown us how small and fragile our home planet truly is. We are learning the imperatives of universal brotherhood and global ecology—learning to think and act as guardians of one tiny blue and green island in the trackless oceans of the universe. This new program will give more people more access to the liberating perspectives of space....

"We must sail sometimes with the wind and sometimes against it," said Oliver Wendell Holmes, "but we must sail, and not drift, nor lie at anchor." So with man's epic voyage into space—a voyage the United States of America has led and still shall lead.¹⁶

It was appropriate to give a name to this new ship of space, and of state. Fletcher, Shapley, and Low had prepared a list that included Pegasus, Hermes, Astroplane, and Skylark. Flanigan passed this list to White House staffers, who picked the name Space Clipper. A draft of Nixon's statement used this name, which resembled Lockheed's Star-Clipper. Nixon himself, however, decided that it would be better to refer to the vehicle in the usual fashion, as the "Space Shuttle." Earlier piloted spacecraft had carried names such as Mercury, Gemini, and Apollo, but the new one would break with this practice. Criticism of Nixon's decision came from the usual sources, with Senator Proxmire issuing a press release: "The President has clearly reordered our priorities. But he has reordered them the wrong way." Senator William Fulbright played the same note: "I believe the shuttle simply cannot rank high on our list of priorities in view of the critical social and economic problems we face." They, however, were merely outriders within a Democratic Party that had long commanded the center of American politics but now was slipping to the left, and whose presidential candidate, George McGovern, would shortly receive one of the worst electoral drubbings ever administered. Though Congress would perform its constitutional role by voting the funding, it would not become an important source of opposition.

16. Low: Personal Notes No. 59, November 28, 1971; No. 62, January 15, 1972; letter, Fletcher to Weinberger, January 4, 1972; memos: David Parker, December 31, 1971; Flanigan to Nixon, January 4, 1972; Nixon, "Statement by the President," January 5, 1972; John Logsdon interview, John Ehrlichman, Santa Fe, May 6, 1983, p. 1; Low, Memo for Record, January 12, 1972. Reprinted in NASA SP-4407, vol. 1, pp. 558-559.

Certainly there was politics aplenty in Nixon's decision. He wanted to help the aerospace industry during the upcoming election year, and the staffer Jonathan Rose, reporting to Flanigan, had been monitoring that industry's unemployment. Fletcher, in a letter to Rose of November 22, had written that an early start on the Shuttle "would lead to a direct employment of 8,800 by the end of 1972, and 24,000 by the end of 1973." Ehrlichman would recall that this was

a very important consideration in Nixon's mind. There are what we call battleground states; they are the pivotal states that control big blocks of electoral votes. So when you look at unemployment numbers, and you key them to the battleground states, the space program has an importance out of proportion to its budget.

The politics also reached a much higher level, touching the matter of presidential decisions that could stand as a legacy, with consequences that would reverberate through coming decades. Theodore White, a chronicler of presidents, had written in 1965 that "on the far edge of the plateau lie problems which we in this decade cannot conceive of as political." These included "the Moon and space. How large a part of American energy should be invested in this exploration with no definable certainty except the certainty that it will change the lives of all our children?"

Weinberger had made an important point in his memo to Nixon of August 12: the United States, as the world's great reserve of strength, could do more than merely add bricks to the welfare state. Such policies might suit the British, of whom the former secretary of state Dean Acheson had said, "Great Britain has lost an Empire and has not yet found a role." They would not suit America.

One could view Nixon's decision as a straightforward exercise in daily work at the White House. Nixon had a strong interest in management; he had set up a staff system, which included a strengthened OMB, that could weigh policy alternatives with considerable effectiveness and present him with well-researched options for his decision. As one Flax Committee staffer put it, "Once they decided to do it, Nixon and Ehrlichman weren't going to argue with NASA whether it ought to be a 60 or a 45-foot-long bay, or 12 or 15 feet. If the head of NASA is telling them that it has to be this size and they want to go ahead with the project, they are going to say go ahead." The demise of

NTOP also helped. While the Space Shuttle stood on its merits and did not simply replace NTOP as a backup, Nixon and Shultz had been prepared to include funding for NTOP initiatives in the FY 1973 budget. When no suitable proposals came forth, that made it easier to shift the putative NTOP funds over to the Shuttle, and to approve a larger Shuttle as well.

Yet while the Shuttle could not match the significance of Nixon's opening to China, it drew on more than good staff work. In Ehrlichman's words, "There wasn't anybody who made those final decisions except Nixon, in this kind of area. Defense, space, certain kinds of domestic problems—he was the final arbiter." The Shuttle carried Nixon's personal stamp because it carried his personal decision.¹⁷

on

e Shuttle on a silver platter, the question of choosing a booster ar that the Shuttle indeed would is conventional first stages were ney could not simply walk away 45,000 pounds; NASA itself had 1 because it offered the potential nch solid rocket motors. Further he OMB from complaining that ssibility with unseemly haste. veight was still open, the basic initive form. During the fall of

1974, when the Mark I/Mark II approach was still in the forefront, the contractors had worked from a Max Faget configuration known as MSC-040A. It elaborated the earlier MSC-040 by adding small liquid-fuel engines for orbital maneuvers, along with thrusters for attitude control that were mounted at the tips of the wings and tail.

The Mark I/Mark II concept, however, with its phased technology, had never been more than an artificial stratagem to reduce peak funding by stretch-

17. *National Journal*, August 19, 1972, p. 1329; John Logsdon interviews: Dave Elliot, p. 4; John Ehrlichman, Santa Fe, May 6, 1983, pp. 9, 32-33; White, 1964, pp. 476, 478; *Oxford*, p. 1; letter, Fletcher to Rose, November 22, 1971. Reprinted in NASA SP-407, vol. 1, pp. 555-558.